## STRUCTURAL DYNAMICS

**3.0 Multi Degree of Freedom (MDOF) System (8 hours)**

3.1 Simple System (Basic concept, uses and limitations)

3.2 Generalized coordinate and reduction of degrees of freedom (Kinematic constraints and static condensation)

3.3 Dynamic equilibrium: Mathematical modeling and dynamic equilibrium equation; influence coefficients (Stiffness influence coefficients, damping influence coefficients and mass influence coefficients)

3.4 Free vibration analysis of undamped system: Eigenvalue problem (natural frequencies and mode shapes)

3.5 Free vibration response of undamped system: Model expansion; Orthogonality conditions; Normalization; Normal coordinates; Uncoupled equations of motion; Mode superposition method

3.6 Modal response analysis of damped systems

3.7 Forced vibration response of damped and undamped system

3.8 Dynamic Analysis of Linear MDOF Systems: Response spectrum analysis; Element forces; Modal contribution factors and minimum number of modes; Base shear; Modal mass

3.9 Evaluation of natural frequencies and mode shapes (Iterative methods): Rayleigh's method; Stodola's method; Holzer method

3.10 Support excitation (Influence vector, synchronous support motion of a planar system and structure with multiple support motions)



## 3.1 Simple Systems

In general, the dynamic response of a structure cannot be described adequately by a SDOF model.

Such behavior can be described only in terms of more than one displacement coordinate; that is, the motion must be represented by more than one degree of freedom.

Multi Degree of Freedom (MDOF) Systems are defined as those requiring two or more coordinates to describe their motion.

It also includes continuous systems which theoretically have an infinite number of freedom.

<u>MDOF system have two groups:</u>

1) Lumped Mass System 2) Continuous system having infinite nodes.

## 3.1 Simple Systems

Consider vibration of a beam. The motion of the beam will be assumed to be defined by the displacements of a set of discrete points of the beam.

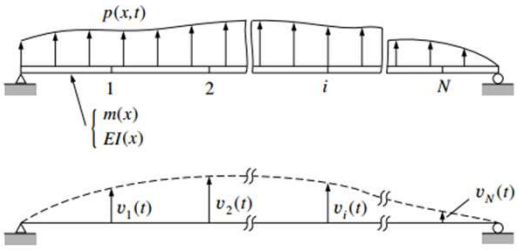
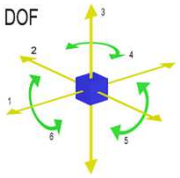



**FIGURE 9-1**
Discretization of a general beam-type structure.

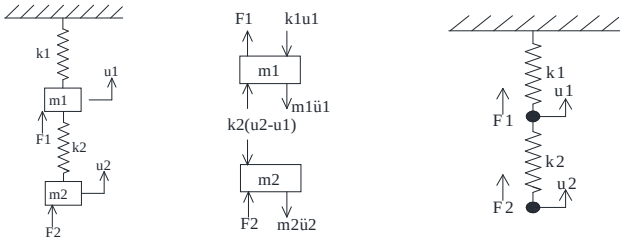
Each nodal point has one degree of freedom but may be three displacement and three rotation. Considering only vertical displacement the beam have 'N' degree of freedom






## 3.1 Simple Systems

Equation of motion for simple undamped MDOF system:




Equation of motion from Newton's 2nd Law and d'Alembert's Principle.

Equating all the forces acting on each mass to zero, we have,

For $m_1$:

$F_1 - m_1\ddot{u}_1 - k_1u_1 + k_2(u_2 - u_1) = 0$ … … (1)

And For $m_2$:

$$F_2 - m_2\ddot{u}_2 - k_2(u_2 - u_1) = 0 \quad \text{——— (20)}$$

writing eqns (19) & (20) in matrix form gives.

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \quad \text{——— (21)}$$

or $[m]\{\ddot{u}\} + [k]\{u\} = \{F\}$

(5) Equation of motion from the stiffness matrix

we can find stiffness matrix as by removing masses.

$$\begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} = \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad \text{——— (22)}$$

Now replacing the masses $m_1$ and $m_2$, the forces $F_1$ and $F_2$ in eqn (22) become, $(F_1 - m_1\ddot{u}_1)$ and $(F_2 - m_2\ddot{u}_2)$ respectively

$$\begin{Bmatrix} F_1 - m_1\ddot{u}_1 \\ F_2 - m_2\ddot{u}_2 \end{Bmatrix} = \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad \text{——— (23)}$$





which can be written as: (8)

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \quad \text{——— (24)}$$

(2nd) Equation of motion from Lagrange's Equations.

If the kinetic energy does not depend upon the displacements of the masses, and damping is ignored then Lagrange's equations can be written as

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial U}{\partial q_i} = Q_i \ (i = 1, 2) \quad \text{——— (25)}$$

where, $T$ = total kinetic energy of the systems

$U$ = total potential energy.

$q_i$ = generalized coordinates

$Q_i$ = generalized forces.

Certain restrictions apply to the choice of these generalized quantities. (a) The coordinates must be independent of each other
(b) The generalized forces, $Q_i$ must do the same work as the actual forces

These requirements are satisfied if we let $q_1 = u_1$, $q_2 = u_2$, $Q_1 = F_1$ and $Q_2 = F_2$.

Again, $T = \frac{1}{2} m_1 \dot{u}_1^2 + \frac{1}{2} m_2 \dot{u}_2^2$ — (26)

and $U = \frac{1}{2} k_1 u_1^2 + \frac{1}{2} k_2 (u_2 - u_1)^2$

$= \frac{1}{2} k_1 u_1^2 + \frac{1}{2} k_2 u_2^2 - k_2 u_1 u_2 + \frac{1}{2} k_2 u_1^2$ — (27)





Then,

$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_1}\right) = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{u}_1}\right) = m_1 \ddot{u}_1$, $\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_2}\right) = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{u}_2}\right) = m_2 \ddot{u}_2$

$\frac{\partial U}{\partial q_1} = \frac{\partial U}{\partial u_1} = k_1 u_1 - k_2 u_2 + k_2 u_1$, $\frac{\partial U}{\partial q_2} = \frac{\partial U}{\partial u_2} = k_2 u_2 - k_2 u_1$

Substituting into eqn (25)

$m_1 \ddot{u}_1 + k_1 u_1 - k_2 u_2 + k_2 u_1 = F_1$
and $m_2 \ddot{u}_2 + k_2 u_2 - k_2 u_1 = F_2$ — (28)

or expressing in matrix form

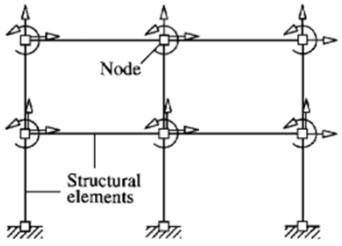
$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix}$$ — (29)

## 3.2 Generalized coordinate and reduction of degree of freedom

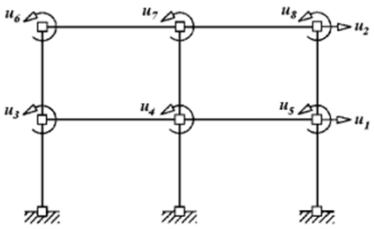
General approach for linear system:

Discretization:

A frame structure can be idealized as assemblage of elements – beams, columns, walls interconnected at nodal points. The displacements of the nodes are the degree of freedom (DOF). The static condensation method is used to eliminate those DOFs of a structure to which zero mass is assigned.



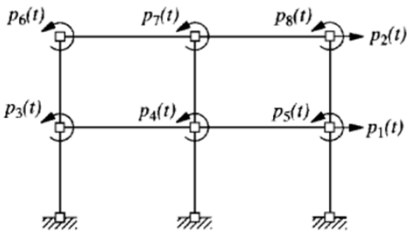

Plane frame with axial deformation - 18 DOFs




Axial deformation neglected - 8 DOFs



## 3.2 Generalized coordinate and reduction of degree of freedom




External dynamic forces

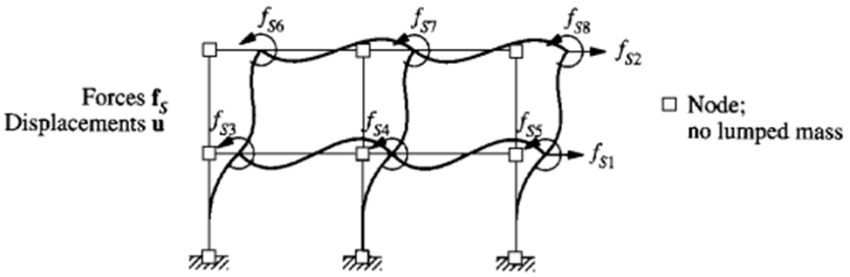
The moments P3(t) to P8(t) are zero in most practical cases.

Elastic Forces:

For linear system, elastic force – displacement relationship can be obtained by the method of superposition and the concept of stiffness influence coefficients.

## 3.2 Generalized coordinate and reduction of degree of freedom

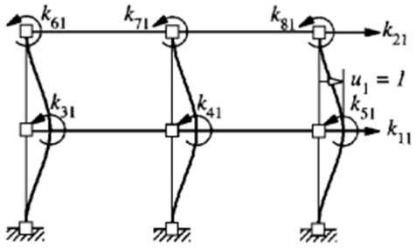
Apply a unit displacement along DOF 'j', holding all other displacements to zero. To maintain this condition, forces must be applied along all DOFs. The stiffness influence coefficient '$k_{ij}$' is the force required along DOF 'i' due to unit displacement at DOF 'j'.



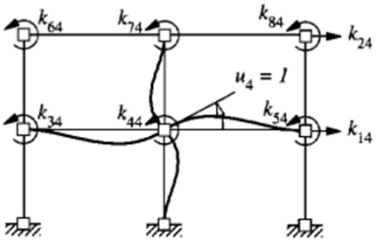

Stiffness component of frame



## 3.2 Generalized coordinate and reduction of degree of freedom




Stiffness influence coefficient for $u_1=1$




Stiffness influence coefficient for $u_4=1$

The force fsi at DOF 'i' associated with displacement $u_j$, j=1 to N is obtained by method of superposition.

$$fs_i = k_{i1}u_1+k_{i2}u_2+.....+k_{ij}u_j+.....k_{iN}u_N \quad ... \quad (30)$$

In matrix form:-

$$\begin{Bmatrix} f_{S1} \\ f_{S2} \\ \vdots \\ f_{SN} \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1j} & \cdots & k_{1N} \\ k_{21} & k_{22} & \cdots & k_{2j} & \cdots & k_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ k_{N1} & k_{N2} & \cdots & k_{Nj} & \cdots & k_{NN} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{Bmatrix} \quad (31)$$

$$\{f_S\} = [k]\{u\}$$

**Damping forces:-**

The energy of a vibrating structure is dissipated can usually be idealized by equivalent viscous damping. With this assumption we relate the external forces $f_{Dj}$ acting on the damping component of the structure to the velocities $\dot{u}_j$.

we impart a unit velocity along DOF j, while the velocities in all other DOFs are kept zero. These velocities will generate internal damping forces that resist the velocities, and external forces





would be necessary to equilibrate these (12) forces.

∴ The damping influence coefficient $C_{ij}$ is the external force in DOF i due to unit velocity in DOF j. The force $f_{Di}$ at DOF i associated with velocities $\dot{u}_j$, j = 1 to N is obtained by superposition.

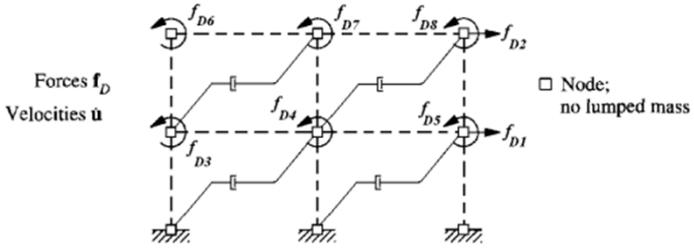



**Figure 9.2.4** Damping component of frame.

## 3.2 Generalized coordinate and reduction of degree of freedom

$$f_{Di} = C_{i1}\dot{u}_1 + C_{i2}\dot{u}_2 + \cdots + C_{ij}\dot{u}_j + \cdots + C_{iN}\dot{u}_N \quad (32)$$

In matrix Form:

$$\begin{Bmatrix} f_{D1} \\ f_{D2} \\ f_{D3} \\ \vdots \\ f_{DN} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1j} & \cdots & C_{1N} \\ C_{21} & C_{22} & \cdots & C_{2j} & \cdots & C_{2N} \\ \vdots & \vdots & & \vdots & & \vdots \\ C_{N1} & C_{N2} & \cdots & C_{Nj} & \cdots & C_{NN} \end{bmatrix} \begin{Bmatrix} \dot{u}_1 \\ \dot{u}_2 \\ \vdots \\ \dot{u}_N \end{Bmatrix} \quad (33)$$

$$\{f_D\} = [C]\{\dot{u}\} \quad (34)$$

It is impractical to compute the coefficients $C_{ij}$ of the damping matrix directly from the dimensions



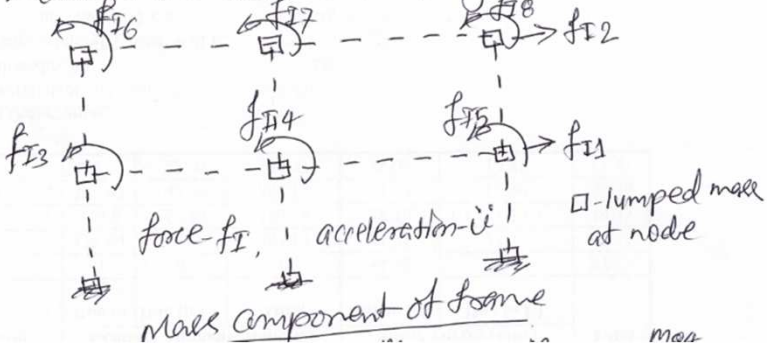
## 3.2 Generalized coordinate and reduction of degree of freedom

of the structure and the sizes of the (13)
structural elements. It is generally specified by numerical values for the damping ratios as for SDOF systems, based on experimental data for similar structures. Methods are available to construct the damping matrix from known damping ratios.

Inertia forces!

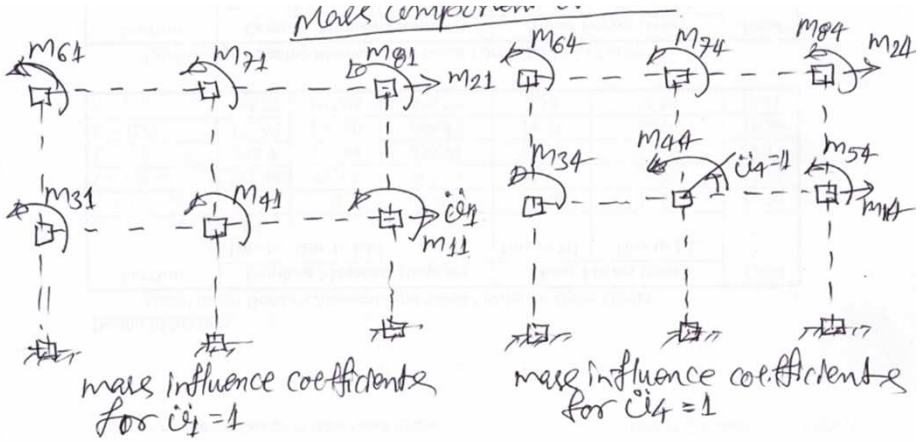
$f_{Ij}$ acting on mass component of the structure to the accelerations $\ddot{v}_j$.

The mass influence coefficient $m_{ij}$ is the external force in DOF $i$ due to unit acceleration along DOF $j$







## 3.2 Generalized coordinate and reduction of degree of freedom

$$f_{I1} = m_{11}\ddot{v}_1 + m_{12}\ddot{v}_2 + \cdots + m_{1j}\ddot{v}_j + \cdots + m_{1N}\ddot{v}_N \quad (34)$$

(35)

In matrix form:-

$$\begin{Bmatrix} f_{I1} \\ f_{I2} \\ \vdots \\ f_{IN} \end{Bmatrix} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1j} & \cdots & m_{1N} \\ m_{21} & m_{22} & \cdots & m_{2j} & \cdots & m_{2N} \\ \vdots & \vdots & & \vdots & & \vdots \\ m_{N1} & m_{2N} & \cdots & m_{Nj} & \cdots & m_{NN} \end{bmatrix} \begin{Bmatrix} \ddot{v}_1 \\ \ddot{v}_2 \\ \vdots \\ \ddot{v}_N \end{Bmatrix} \quad (36)$$

$$\{f_I\} = [m]\{\ddot{v}\} \quad (37)$$

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## 3.3 Dynamic equilibrium

The equation of motion of the system can be formulated by expressing the equilibrium of the effective forces associated with each of its degrees of freedom.

In general four types of forces will be involved at any point:

1) the externally applied load - pi(t)
2) Inertia force – $f_I(i)$,
3) damping force – $f_D(i)$, and
4) elastic force – $f_S(i)$.

Thus for each of the several degrees of freedom the dynamic equilibrium may be expressed as

$$\begin{aligned} f_{I1} + f_{D1} + f_{S1} &= p_1(t) \\ f_{I2} + f_{D2} + f_{S2} &= p_2(t) \quad \ldots \quad \ldots (1) \\ f_{I3} + f_{D3} + f_{S3} &= p_3(t) \end{aligned}$$

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## 3.3 Dynamic Equilibrium

$$f_{I1} + f_{D1} + f_{S1} = P_{1(t)} - for\ Node - 1$$
$$f_{I2} + f_{D2} + f_{S2} = P_{2(t)} - for\ Node - 2 \quad \ldots \quad \ldots (1)$$
$$f_{I3} + f_{D3} + f_{S3} = P_{3(t)} - for\ Node - 3$$
$$- - - - - - - - - - - - - - - - - - - - - - - - - -$$
$$f_{IN} + f_{DN} + f_{SN} = P_{N(t)} - for\ Node - N$$

When force vectors are presented in matrix form:

$$\{f_I\} + \{f_D\} + \{f_S\} = \{P_{(t)}\} \quad \ldots \quad \ldots (2)$$

OR $f_I + f_D + f_S = P_{(t)}$

Which is multi degree of freedom equation.



## 3.3 Dynamic equilibrium

Here, Each type of force component developed at point 1, depends upon the displacement components developed at all points of the structure (2) As like,

$$f_{S1} = k_{11}u_1 + k_{12}u_2 + k_{13}u_3 + \cdots + k_{1N}u_N$$

Similarly the elastic force corresponding to the degree of freedom $u_2$ is

$$f_{S2} = k_{21}u_1 + k_{22}u_2 + k_{23}u_3 + \cdots + k_{2N}u_N \quad (3)$$

in general,

$$f_{Si} = k_{i1}u_1 + k_{i2}u_2 + k_{i3}u_3 + \cdots + k_{iN}u_N$$

In these expressions it has been tacitly assumed that the structural behaviour is linear, so that the principle of superposition applies.

## 3.3 Dynamic equilibrium

The coefficients $K_{ij}$ are called stiffness influence coefficients.

$K_{ij}$ = force corresponding to coordinate $i$ due to a unit displacement of coordinate $j$

In matrix form the complete set of elastic force relationship may be written as:

$$\begin{Bmatrix} f_{S1} \\ f_{S2} \\ \vdots \\ f_{Si} \\ \vdots \\ f_{SN} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} & K_{13} & \cdots & K_{1j} & \cdots & K_{1N} \\ K_{21} & K_{22} & K_{23} & \cdots & K_{2j} & \cdots & K_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ K_{i1} & K_{i2} & K_{i3} & \cdots & K_{ij} & \cdots & K_{iN} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ K_{N1} & K_{N2} & K_{N3} & \cdots & K_{Nj} & \cdots & K_{NN} \end{bmatrix} \begin{Bmatrix} v_1 \\ v_2 \\ \vdots \\ v_i \\ \vdots \\ v_N \end{Bmatrix} \quad \text{---(4)}$$

or symbolically,

$$\{f_S\} = [K]\{v\} \quad \text{---(5)}$$



## 3.3 Dynamic equilibrium

Here, $[K]$ should be symetrical. i.e. $K_{ij} = K_{ji}$

To generate $i$th column, we apply unit displacement at $i$th position.

Damping force:

$$\begin{Bmatrix} f_{D1} \\ f_{D2} \\ \vdots \\ f_{Di} \\ \vdots \\ f_{DN} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots & C_{1i} & \cdots & C_{1N} \\ C_{21} & C_{22} & C_{23} & \cdots & C_{2i} & \cdots & C_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ C_{i1} & C_{i2} & C_{i3} & \cdots & C_{ii} & \cdots & C_{iN} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ C_{N1} & C_{N2} & C_{N3} & \cdots & C_{Ni} & \cdots & C_{NN} \end{bmatrix} \begin{Bmatrix} \dot{v}_1 \\ \dot{v}_2 \\ \vdots \\ \dot{v}_i \\ \vdots \\ \dot{v}_N \end{Bmatrix} \quad \text{---(6)}$$

$\dot{v}_i$ = velocity of the $i$ displacement coordinate

$C_{ij}$ = damping influence coefficients.

= force corresponding to coordinate $i$ due to unit velocity of coordinate $j$

## 3.3 Dynamic equilibrium

$$\therefore \{f_D\} = [C]\{\dot{v}\} \quad \text{---(7)}$$

$[C]$ = damping matrix

Inertial forces:-

$$\begin{Bmatrix} f_{I1} \\ f_{I2} \\ \cdots \\ f_{Ii} \\ \cdots \\ f_{IN} \end{Bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & \cdots & m_{1i} & \cdots & m_{1N} \\ m_{21} & m_{22} & m_{23} & \cdots & m_{2i} & \cdots & m_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ m_{i1} & m_{i2} & m_{i3} & \cdots & m_{ii} & \cdots & m_{iN} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ m_{N1} & m_{N2} & m_{N3} & \cdots & m_{Ni} & \cdots & m_{NN} \end{bmatrix} \begin{Bmatrix} \ddot{v}_1 \\ \ddot{v}_2 \\ \cdots \\ \ddot{v}_i \\ \cdots \\ \ddot{v}_N \end{Bmatrix} \quad \text{---(8)}$$

$\ddot{v}_i$ = acceleration of the $i$ displacement coordinate

$m_{ij}$ = mass influence coefficients

= force corresponding to coordinate $i$ due to unit acceleration of coordinate $j$



## 3.3 Dynamic equilibrium

$$\therefore \{f_I\} = [m]\{\ddot{v}\} \quad \text{---(9)}$$

$[m]$ = mass matrix

Substituting in eq$^n$ (2) we get,

$$[m]\{\ddot{v}\} + [C]\{\dot{v}\} + [K]\{v\} = \{P(t)\} \quad \text{---(10)}$$

Dynamic equation

This equation is the MDOF system having N degree of freedom.

For a system in which only translational degree of freedom are defined, the lumped-mass matrix has a diagonal form.

$$[m] = \begin{bmatrix} m_1 & 0 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & m_2 & 0 & \cdots & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & m_i & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & \cdots & m_N \end{bmatrix}$$

~~hence, $m_{22} = m_2$~~

---(11)

## 3.3 Dynamic equilibrium

The off-diagonal terms $m_{ij}$ of this matrix vanish because an acceleration of any mass point produces an inertial force at that point only.

$$\therefore m_{ij} = m_i$$

If more than one translational degree of freedom is specified at any nodal point, the same point mass will be associated with each degree of freedom. On the other hand the mass associated with any rotational degree of freedom will be zero because of assumption that the mass is lumped at point which have no rotational inertia. If a rigid mass having a finite rotational inertia is associated with a rotational degree of freedom the diagonal mass coefficient for that degree of freedom would be the rotational inertia of the mass.
In general the lumped mass matrix is a diagonal matrix which will include zero diagonal elements for the rotational degree of freedom.

Consistent-mass matrix : Point mass include rotational effect.



## 3.4 Free Vibration analysis of undamped system: Eigen value problem

<u>Eigenvalues and Eigenvectors:</u>
Eigenvalues and eigenvectors are mathematical concepts, and their use is not confined to vibration theory. The word incidentally are derived from the German word “eigen” meaning “own”. So the eigenvalues are its own values and the eigenvectors are its own vectors.

Eigenvalues and eigenvectors can be real or complex, and this is determined entirely by the damping in the system.

System with no damping have real eigenvalues and eigenvectors.

Suppose we wish to find the eigenvalues and eigenvectors of the general undamped system.

## 3.4 Free Vibration analysis of undamped system: Eigen value problem

$$[m]\{\ddot{u}\} + [k]\{u\} = \{P(t)\} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

Since the eigenvalues and eigenvectors are unaffected by the applied forces, they can be obtained from the homogeneous equations given by omitting the external forces:

$$[m]\{\ddot{u}\} + [k]\{u\} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Or,
$$\begin{bmatrix} m_{11} & m_{21} & \ldots & \ldots & m_{1n} \\ m_{21} & m_{22} & \ldots & \ldots & m_{2n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ m_{n1} & m_{n2} & \ldots & \ldots & mnn \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ldots \\ \ldots \\ \ddot{u}_n \end{Bmatrix} + \begin{bmatrix} k_{11} & k_{12} & \ldots & \ldots & k_{1n} \\ k_{21} & k_{22} & \ldots & \ldots & k_{2n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ k_{n1} & kn_2 & \ldots & \ldots & knn \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ \ldots \\ \ldots \\ u_n \end{Bmatrix} = 0$$

The responses at all the points $u_1$, $u_2$, ... etc can be considered as horizontal or vertical projections of a rotating, complex unit vector, $e^{iwt}$ multiplied by the real constants $\hat{u}_1$, $\hat{u}_2$ etc.



## 3.4 Free Vibration analysis of undamped system: Eigen value problem

Then the responses of all the points $u_1$, $u_2$, etc are given by,

$$\{u\} = \begin{Bmatrix} u_1 \\ u_2 \\ \ldots \\ \ldots \\ u_n \end{Bmatrix} = \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \ldots \\ \ldots \\ \hat{u}_n \end{Bmatrix} e^{i\omega t} = \{\hat{u}\} e^{i\omega t} \ldots \quad \ldots \quad (3)$$

Differentiating twice w.r.t. 't'

$$\{\ddot{u}\} = \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ldots \\ \ldots \\ \ddot{u}_n \end{Bmatrix} = -\omega^2 \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \ldots \\ \ldots \\ \hat{u}_n \end{Bmatrix} e^{i\omega t} = -\omega^2 \{\hat{u}\} e^{i\omega t} \ldots \quad \ldots \quad (4)$$

Substituting in equation no (2), we get,

## 3.4 Free Vibration analysis of undamped system: Eigen value problem

$$-\omega^2 \begin{bmatrix} m_{11} & m_{21} & \dots & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & \dots & m_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ m_{n1} & m_{n2} & \dots & \dots & mnn \end{bmatrix} \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \dots \\ \dots \\ \hat{u}_n \end{Bmatrix} + \begin{bmatrix} k_{11} & k_{12} & \dots & \dots & k_{1n} \\ k_{21} & k_{22} & \dots & \dots & k_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ k_{n1} & kn_2 & \dots & \dots & knn \end{bmatrix} \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \dots \\ \dots \\ \hat{u}_n \end{Bmatrix} = 0 \quad \dots \quad \dots \quad (5)$$

Or, $\{-\omega^2[m] + [k]\{\hat{u}\} = 0 \quad \dots \quad \dots . \quad (6)$

Which is eigenvalue problem.

In general, by analogy with the behaviour of SDOF system, it will be assumed that the free vibration of motion is simple harmonic, which may be expressed for MDOF system as

$$\{u_t\} = \{\hat{u}\}.Sin(\omega t + \theta) \quad \dots . \quad \dots . . (7)$$

[As like in SDOF system, ut = ρ.Sin(ωt+φ) ]
$\{\hat{u}\}$ denotes shape of the system, does not change with time, only magnitude varies.



## 3.4 Free Vibration analysis of undamped system

θ - phase angle

Second time derivative of equation (7) gives,

$\{\ddot{u}_t\} = -\omega^2\{\hat{u}\}.Sin(\omega t + \theta)$ = $-\omega^2\{u_t\}$ … …. (8)

Substituting equations (7) & (8) in equation (2), we get,

$[m]\{-\omega^2\{\hat{u}\}.Sin(\omega t + \theta)\}+[k]\{\{\hat{u}\}.Sin(\omega t + \theta)\} = 0$

$\left[[k] - \omega^2[m]\right]\{\hat{u}\}.Sin(\omega t + \theta) = 0$ … … (9)

Omitting arbitrary Sine terms,

$\left[[k] - \omega^2[m]\right]\{\hat{u}\} = 0$ … … (10)

The above equation is called eigenvalue or characteristic value problem.
$\omega^2$ - eigenvalues or characteristics value, and
$\{\hat{u}\}$ - shape of the vibrating system known as eigenvector or mode shape.

## 3.4 Free Vibration analysis of undamped system

It can be shown by Cramer's rule that the solution of this set of simultaneous equations is of the form of

$$\{\hat{u}\} = \frac{0}{|[k]-\omega^2[m]|} \quad \ldots \quad \ldots \quad (11)$$

Non trivial solution is possible only when determinant $[k] - \omega^2[m]$ vanishes. i. e.

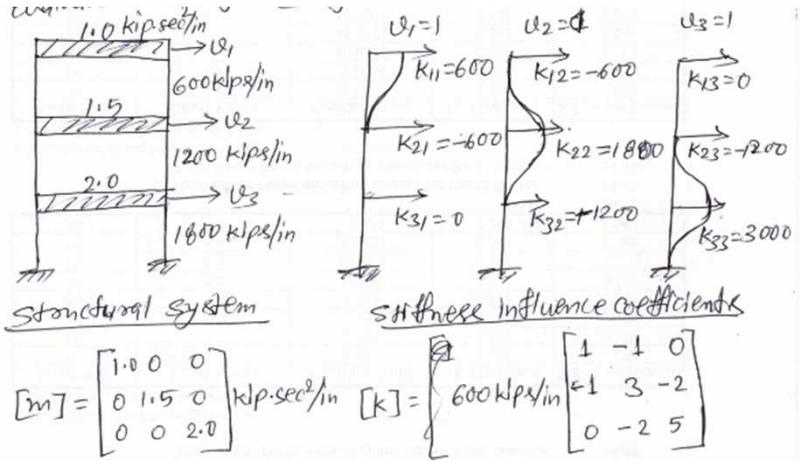
$$|[k] - \omega^2[m]| = 0 \quad \ldots \quad \ldots \quad (12)$$

Above equation is known as frequency equation of the system.



## 3.4 Free Vibration analysis of undamped system

Ex 1: Analysis the given system for vibration frequencies.




$$[m] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} \text{kip.sec}^2/\text{in} \qquad [k] = 600 \text{ kips/in} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 5 \end{bmatrix}$$

## 3.4 Free Vibration analysis of undamped system

$$\therefore [k] - \omega^2[m] = 600 \text{ kips/in} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 5 \end{bmatrix} - \omega^2 \text{ kips-sec}^2/\text{in} \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} \text{ kips/in}$$

Substitute, $\omega^2 = 600B$, then,

$$= 600 \begin{bmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 5 \end{bmatrix} - 600B \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} = 600 \begin{bmatrix} 1-B & -1 & 0 \\ -1 & 3-1.5B & -2 \\ 0 & -2 & 5-2B \end{bmatrix} \quad \text{---(a)}$$



## 3.4 Free Vibration analysis of undamped system

The frequencies of the frame are given by the condition that $\Delta = 0$,

$$\therefore \begin{vmatrix} 1-B & -1 & 0 \\ -1 & 3-1.5B & -2 \\ 0 & -2 & 5-2B \end{vmatrix} = 0$$

$$\text{or } (1-B)[(3-1.5B)(5-2B) - 4] + 1 \times [-1(5-2B) - 0] = 0$$

$$\text{or } (1-B)(15 - 6B - 7.5B + 3B^2 - 4) + (-5 + 2B) = 0$$

$$\text{or } (1-B)(11 - 13.5B + 3B^2) + (-5 + 2B) = 0$$

$$\text{or } 11 - 13.5B + 3B^2 - 11B + 13.5B^2 - 3B^3 - 5 + 2B = 0$$

$$\text{or } -3B^3 + 16.5B^2 - 22.5B + 6 = 0$$

$$\text{or } B^3 - 5.5B^2 + 7.5B - 2 = 0$$

Solving

Q $\beta^3 - 5.515\beta^2 + 4.515\beta - 1 = 0$

The three roots of this equation may be obtained by solving or by trial and error.

$\beta_1 = 0.3515, \quad \beta_2 = 1.6066, \quad \beta_3 = 3.5420$

$$\therefore \{\omega^2\} = \begin{Bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \end{Bmatrix} = 600 \begin{Bmatrix} 0.3515 \\ 1.6066 \\ 3.5420 \end{Bmatrix} = \begin{Bmatrix} 210.88 \\ 963.96 \\ 2125.20 \end{Bmatrix}$$

$$\therefore \{\omega\} = \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} = \begin{Bmatrix} 14.522 \\ 31.048 \\ 46.100 \end{Bmatrix} \text{ rad/sec.}$$

... of the undamp



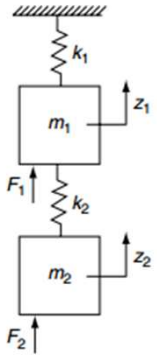


Ex.2 Find the eigenvalues and eigenvectors of the undamped system shown with, $m_1$=1kg; $m_2$=2kg; $k_1$=10N/m; $k_2$=10N/m.

Scale the eigenvectors so that the largest absolute element in each column is set to unity.

a) Demonstrate that a transformation to modal coordinates using the eigenvectors as modes enables the equations to be written as uncouplled SDOF systems.

b) Rescale the eigenvectors so that the mass matrix in normal mode coordinates, is a unit matrix.



Solution: Part (a)

The equation of motion for this system is

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \quad \dots \quad \dots \quad (1)$$

Inserting given numerical value, we get

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} 20 & -10 \\ -10 & 10 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \quad \dots \quad \dots \quad (2)$$

Omitting the applied forces F1 and F2 gives the homogeneous equations from which the eigenvalues and eigenvectors can be found.

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} 20 & -10 \\ -10 & 10 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad \dots \quad \dots \quad (3)$$

Arranging in to form of $([k]-\omega^2[m])\{\hat{u}\} = 0$, we have,

$$\left( \begin{bmatrix} 20 & -10 \\ -10 & 10 \end{bmatrix} - \omega 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \right) \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad \dots \quad \dots \quad (4)$$





Say $\omega 2 = \lambda$

$$\begin{bmatrix} 20-\lambda & -10 \\ -10 & 10-2\lambda \end{bmatrix} \begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad \dots \quad \dots \quad (5)$$

Two eigenvalues, $\lambda 1$ and $\lambda 2$ are now found by writing the matrix in equation (5) as a determinant and equating it to zero.

$$\begin{vmatrix} 20-\lambda & 10 \\ -10 & 10-2\lambda \end{vmatrix} = 0$$

$(20-\lambda)(10-2\lambda) - 100 = 0$

$200 - 40\lambda - 10\lambda + 2\lambda^2 - 100 = 0$

$\lambda^2 - 25\lambda + 50 = 0$

$\lambda = 2.192$ and $22.808$

Therefore, two natural frequencies are:

$\omega 1 = \sqrt{\lambda_1} = 1.480$ rad/sec and $\omega 2 = \sqrt{\lambda_2} = 4.776$ rad/sec

The two eigenvectors can now be found by substituting the eigenvalues, one at a time into equation no (5)

## 3.4 Free Vibration analysis of undamped system

$(20-\lambda_1)\hat{u}_1 - 10\hat{u}_2 = 0$

$\frac{\hat{u}_1}{\hat{u}_2} = \frac{10}{20-\lambda}$ or $\frac{\hat{u}_1}{\hat{u}_2} = \frac{10-2\lambda}{10}$

Taking $\lambda = \lambda_1 = 2.192$, $\left(\frac{\hat{u}1}{\hat{u}2}\right)_1 = 0.5615$

Taking $\lambda = \lambda_2 = 22.808$, $\left(\frac{\hat{u}1}{\hat{u}2}\right)_2 = -3.5613$

The eigenvectors can now be called "normal modes", and it is usual to write them as vectors,

$\{\phi\}_1 = \begin{Bmatrix}\hat{u}_1\\ \hat{u}_2\end{Bmatrix}_1$ and $\{\phi\}_2 = \begin{Bmatrix}\hat{u}_1\\ \hat{u}_2\end{Bmatrix}_2$ … … (8)

Using the first of the standard methods for normalizing the eigenvectors (making the largest absolute element in each column to unity) these are:

$\{\phi\}_1 = \begin{Bmatrix}0.5615\\ 1\end{Bmatrix}_1$ and $\{\phi\}_2 = \begin{Bmatrix}1\\ -0.2808\end{Bmatrix}_2$



## 3.4 Free Vibration analysis of undamped system

Analysis of vibration mode shapes:

We have, $([k] - \omega 2[m])\{\hat{u}\} = 0$ … … (16)

Where frequency vector, $\omega = \begin{Bmatrix}\omega_1\\ \omega_2\\ \dots\\ \dots\\ \omega_N\end{Bmatrix}$

Substituting, $[k] - \omega^2[m] = [\tilde{E}^{(n)}]$

$[\tilde{E}^{(n)}]\{\hat{u}\} = 0$ … … (38)

$[\tilde{E}^{(n)}]$ depends on "ω", it is different for each mode.

## 3.4 Free Vibration analysis of undamped system

Equation (16) is satisfied identically, and the amplitude of vibration is indeterminate. However, the shape of the vibrating system can be determined by solving for all displacements in terms of any one coordinate.

For this purpose one (first value) is assumed. i.e.

$\hat{U}_{1n} = 1$, then

$$\begin{Bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \dots \\ \dots \\ \hat{u}_n \end{Bmatrix} \cong \begin{Bmatrix} 1 \\ \hat{u}_{n2} \\ \hat{u}_{n3} \\ \dots \\ \dots \\ \hat{u}_{nn} \end{Bmatrix} \quad \dots \quad \dots \quad (39)$$



## 3.4 Free Vibration analysis of undamped system

In expanded form of equation (38) can be written as:

$$\begin{bmatrix} e_{11}^{(n)} & e_{12}^{(n)} & e_{13}^{(n)} & \dots & \dots & e_{1N}^{(n)} \\ e_{21}^{(n)} & e_{22}^{(n)} & e_{23}^{(n)} & \dots & \dots & e_{2N}^{(n)} \\ e_{31}^{(n)} & e_{32}^{(n)} & e_{33}^{(n)} & \dots & \dots & e_{3N}^{(n)} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ e_{n1}^{(n)} & e_{n2}^{(n)} & e_{n3}^{(n)} & \dots & \dots & e_{nN}^{(n)} \end{bmatrix} \begin{Bmatrix} 1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \dots \\ \hat{u}_{NN} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{Bmatrix} \quad \dots \quad \dots \quad (40)$$

Partitioning the matrix with unknown elements

$$\begin{bmatrix} e_{11}^{(n)} & \tilde{E}_{10}^{(n)} \\ \tilde{E}_{01}^{(n)} & \tilde{E}_{00}^{(n)} \end{bmatrix} \begin{Bmatrix} 1 \\ \{\hat{u}_{0N}\} \end{Bmatrix} = \begin{Bmatrix} 0 \\ \{0\} \end{Bmatrix} \quad \dots \quad \dots \quad (41)$$

Or, $\left[\tilde{E}_{01}^{(n)}\right] + \left[\tilde{E}_{00.}^{(n)}\right]\{\hat{u}_{0N}\} = \{0\}$ … … (42), and

$e_{11}^{(n)} + \left[\tilde{E}_{10}^{(n)}\right]\{\hat{u}_{oN}\} = 0$ … … (43)

Equation (42) can be solved simultaneously for displacement amplitudes

$$\{\hat{u}_{0N}\} = -\left[\tilde{E}_{00}^{(n)}\right]^{-1}\left[\tilde{E}_{01}^{(n)}\right] \quad \dots \quad \dots . \quad (44)$$

But equation (43) is redundant, and satisfied identically.

For convenience the vector is usually expressed in dimension less form by dividing all the components by one reference component (usually the largest). The resulting vector is called the $n^{th}$ mode shape $\phi_n$, thus,

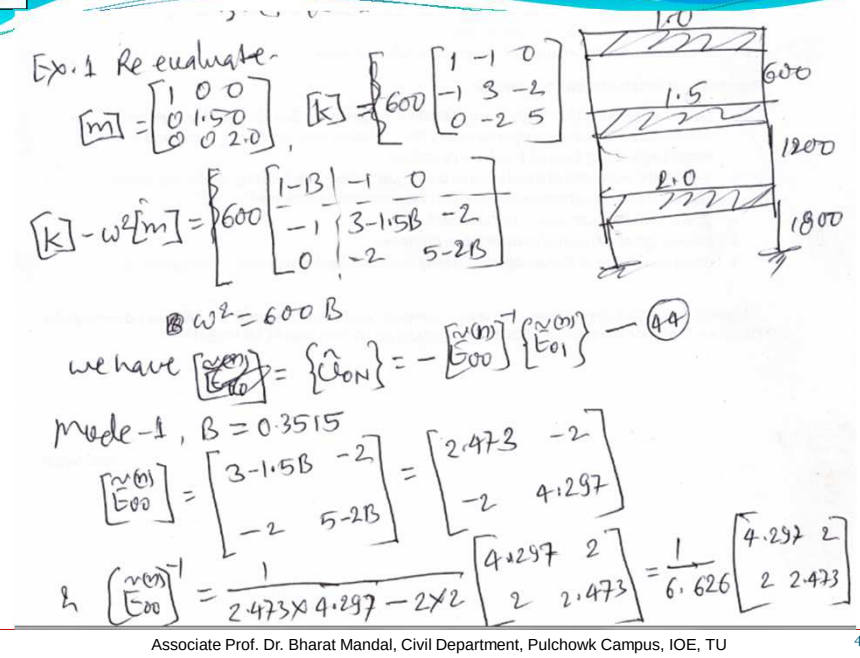
$$\phi_n = \begin{Bmatrix} \phi_{1n} \\ \phi_{2n} \\ \phi_{3n} \\ \dots \\ \dots \\ \phi_{Nn} \end{Bmatrix} \cong \frac{1}{\hat{u}_{kn}} \begin{Bmatrix} 1 \\ \hat{u}_{2n} \\ \hat{u}_{3n} \\ \dots \\ \dots \\ \hat{u}_{Nn} \end{Bmatrix} \quad \dots \quad \dots \quad (45)$$

Where, $\hat{u}_{kn}$ is the reference component taken as the first component, here k = 1

$\therefore$ $([k]-\omega^2[m])\phi_n = 0$







## 3.4 Free Vibration analysis of undamped system

Mode shape vector;

$$\{\phi\}_1 = \begin{Bmatrix}\phi_{11}\\ \phi_{21}\\ \phi_{31}\end{Bmatrix} = \begin{Bmatrix}1\\ \phi_{21}\\ \phi_{31}\end{Bmatrix} \quad \because \tilde{\omega}_1 = 1$$

$$\therefore \begin{Bmatrix}\tilde{\omega}_{21}\\ \tilde{\omega}_{31}\end{Bmatrix} = \begin{Bmatrix}\phi_{21}\\ \phi_{31}\end{Bmatrix} = -\left[\tilde{E}_{00}^{(1)}\right]^{-1}\left\{\tilde{E}_{01}^{(1)}\right\} = -\frac{1}{6.626}\begin{bmatrix}4.297 & 2\\ 2 & 2.473\end{bmatrix}\begin{Bmatrix}-1\\ 0\end{Bmatrix}$$

$$= -\frac{1}{6.626}\begin{Bmatrix}-4.297\\ -2\end{Bmatrix} = \begin{Bmatrix}0.649\\ 0.302\end{Bmatrix}$$

$$\therefore \{\phi\}_1 = \begin{Bmatrix}\phi_{11}\\ \phi_{21}\\ \phi_{31}\end{Bmatrix} = \begin{Bmatrix}1\\ 0.649\\ 0.302\end{Bmatrix}$$

mode-2 $\quad B = 1.6066$

$$\left[\tilde{E}_{00}^{(1)}\right] = \begin{bmatrix}3-1.5\times1.6066 & -2\\ -2 & 5-2\times1.6066\end{bmatrix} = \begin{bmatrix}0.5901 & -2\\ -2 & 1.7868\end{bmatrix}$$

$$\left[\tilde{E}_{00}^{(1)}\right]^{-1} = \frac{1}{-2.9456}\begin{bmatrix}1.7868 & +2\\ +2 & 0.5901\end{bmatrix}$$



## 3.4 Free Vibration analysis of undamped system

$$\begin{Bmatrix}\phi_{22}\\ \phi_{32}\end{Bmatrix} = -\left[\tilde{E}_{00}^{(1)}\right]^{-1}\left\{\tilde{E}_{01}^{(1)}\right\} = \frac{-1}{-2.9456}\begin{bmatrix}1.7868 & 2\\ 2 & 0.5901\end{bmatrix}\begin{Bmatrix}-1\\ 0\end{Bmatrix} = \begin{Bmatrix}0.607\\ 0.679\end{Bmatrix}$$

$$\therefore \{\phi_2\} = \begin{Bmatrix}\phi_{21}\\ \phi_{22}\\ \phi_{23}\end{Bmatrix} = \begin{Bmatrix}1.000\\ -0.607\\ -0.679\end{Bmatrix}$$

mode-3 $\quad B = 3.542$

$$\left[\tilde{E}_{00}^{(1)}\right] = \begin{bmatrix}3-5.313 & -2\\ -2 & 5-7.084\end{bmatrix} = \begin{bmatrix}-2.313 & -2\\ -2 & -2.084\end{bmatrix}$$

$$\left[\tilde{E}_{00}^{(1)}\right]^{-1} = \frac{1}{+0.820}\begin{bmatrix}-2.084 & 2\\ 2 & -2.313\end{bmatrix}$$

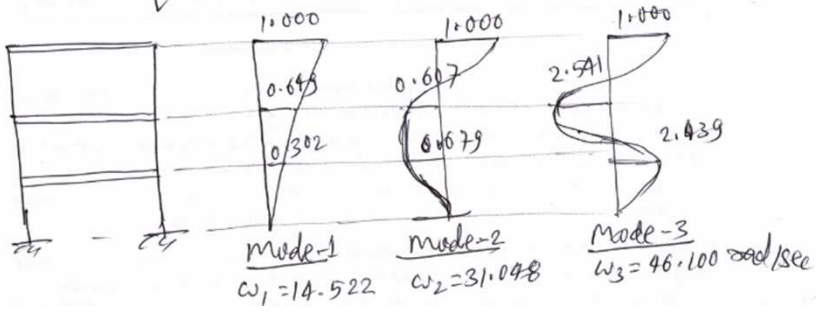
$$\begin{Bmatrix}\phi_{32}\\ \phi_{33}\end{Bmatrix} = -\left[\tilde{E}_{00}^{(1)}\right]^{-1}\left\{\tilde{E}_{01}^{(1)}\right\} = \frac{-1}{+0.820}\begin{bmatrix}-2.084 & 2\\ 2 & -2.313\end{bmatrix}\begin{Bmatrix}-1\\ 0\end{Bmatrix} = \begin{Bmatrix}-2.541\\ +2.439\end{Bmatrix}$$

$$\therefore \{\phi_3\} = \begin{Bmatrix}\phi_{31}\\ \phi_{32}\\ \phi_{33}\end{Bmatrix} = \begin{Bmatrix}1.000\\ -2.541\\ 2.439\end{Bmatrix}$$

## 3.4 Free Vibration analysis of undamped system

$$[\phi] = [\{\phi_1\}, \{\phi_2\}, \{\phi_3\}]$$

$$= \begin{bmatrix} 1.000 & 1.000 & 1.000 \\ 0.648 & -0.607 & -2.541 \\ 0.302 & -0.679 & 2.439 \end{bmatrix}$$





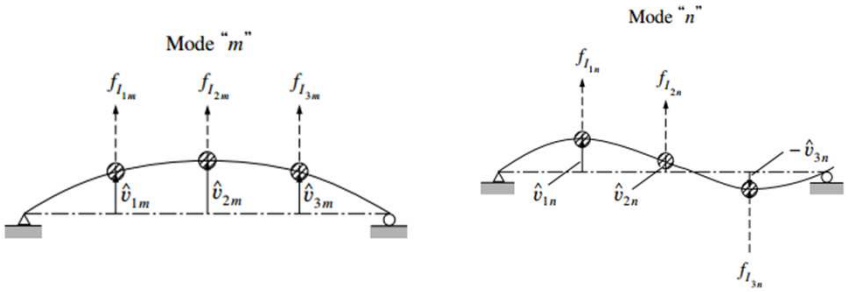

## 3.5 Free Vibration response of undamped system: Orthogonality conditions

Betty's Law:

Work done by force at node 1 because of unit displacement at node 2 = work done by force at node 2 due to unit displacement at node 1.

Virtual work = Real force x virtual displacement
= virtual force x real displacement

Mass system:- (Undamped free vibration)



## 3.5 Free Vibration response of undamped system: Orthogonality conditions

We have,
$[m]\{\ddot{u}\} + [k]\{u\} = \{0\}$

$[k]\{u\} = -[m]\{\ddot{u}\}$

We also have, $([k] - \omega_n^2[m])\{\hat{u}_n\} = \{0\}$

Or, $[k]\{\hat{u}_n\} = \omega_n^2[m]\{\hat{u}_n\}$ ... ... 46

Or, $[k]\{\phi_n\} = \omega_n^2[m]\{\phi_n\}$

$[k]\{\hat{u}_n\}$: Elastic resisting force vector – $\{f_s\}$

$\omega_n^2[m]\{\hat{u}_n\}$ : Applied inertia load vector – $\{f_I\}$



## 3.5 Free Vibration response of undamped system: Orthogonality conditions

Since mode 'm' and mode 'n' are applied in the same system, but different force and displacement, therefore Betty's Law can be applied.

$\{f_{Im}\}^T\{\hat{u}_n\} = \{f_{In}\}^T\{\hat{u}_m\}$ ... ... 47

$(\omega_m^2[m]\{\hat{u}_m\})^T\{\hat{u}_n\} = (\omega_n^2[m]\{\hat{u}_n\})^T\{\hat{u}_m\}$

Reversal law:

$([a][b][c][d])^T = [d]^T[c]^T[b]^T[a]^T$

$([a][b][c][d])^{-1} = [d]^{-1}[c]^{-1}[b]^{-1}[a]^{-1}$

## 3.5 Free Vibration response of undamped system: Orthogonality conditions

Using Reversal law:

$\omega_m^2 \{\hat{u}_m\}^T [m]^T \{\hat{u}_n\} = \omega_n^2 \{\hat{u}_n\}^T [m]^T \{\hat{u}_m\}$ ... ... 48

$\omega_m^2 \{\hat{u}_m\}^T [m] \{\hat{u}_n\} = \omega_n^2 \{\hat{u}_n\}^T [m] \{\hat{u}_m\}$ ... ... 49

since $[m]^T = [m]$

(It can be noted that the matrix products in above Eq. are scalars and can be transposed arbitrarily.)

$\omega_m^2 \{\hat{u}_m\}^T [m] \{\hat{u}_n\} = \omega_n^2 \{\hat{u}_m\}^T [m] \{\hat{u}_n\}$

$(\omega_m^2 - \omega_n^2) \{\hat{u}_m\}^T [m] \{\hat{u}_n\} = 0$ ... ... 50

Since $\omega_m \neq \omega_n$ $\therefore \{\hat{u}_m\}^T [m] \{\hat{u}_n\} = 0$ ... ... 51

**It is known as first orthogonal condition.**



## 3.5 Free Vibration response of undamped system: Orthogonality conditions

Pre multiplying equation 46 by $\{\hat{u}_m\}^T$ we get,

$[k] \{\hat{u}_n\} = \omega_n^2 [m] \{\hat{u}_n\}$ ... ... 46

$\{\hat{u}_m\}^T [k] \{\hat{u}_n\} = \omega_n^2 \{\hat{u}_m\}^T [m] \{\hat{u}_n\}$

From equation 51,

$\{\hat{u}_m\}^T [k] \{\hat{u}_n\} = 0$ ... ... ... 52

**It is known as second orthogonal condition.**

We can write,

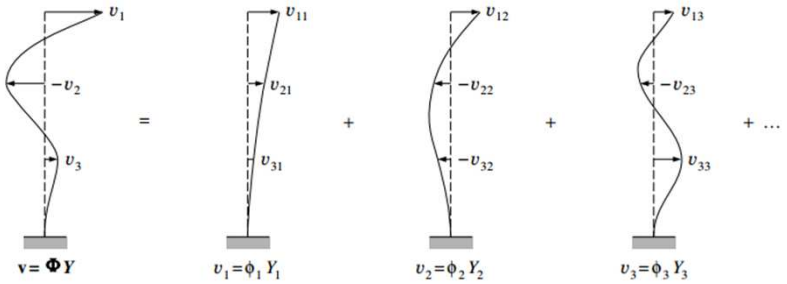
$\{\phi\}_m^T [m] \{\phi\}_n = 0$ , $m \neq n$

$\{\phi\}_m^T [k] \{\phi\}_n = 0$ , $m \neq n$ } ... 53

## 3.5 Free Vibration response of undamped system: Modal superposition method

Analysis of dynamic response using superposition:

Consider, the cantilever column shown in figure for which the deflected shape is expressed in terms of translational displacements at three levels.






## 3.5 Free Vibration response of undamped system: Modal superposition method

Any displacement vector u (static or dynamic) for this structure can be developed by superposing suitable amplitudes of the normal modes as shown.

For any modal component $u_n$, the displacements are given by the product of the mode-shape vector ϕn and the modal amplitude *Y*n; thus

$$\{u_n\} = [\phi_n]\, y_n$$

Or,

$$\begin{Bmatrix} u_{11} \\ u_{21} \\ u_{31} \end{Bmatrix} = \begin{Bmatrix} \emptyset_{11} \\ \emptyset_{21} \\ \emptyset_{31} \end{Bmatrix} y_1 = \begin{Bmatrix} \emptyset_{11} \cdot y_1 \\ \emptyset_{21} \cdot y_1 \\ \emptyset_{31} \cdot y_1 \end{Bmatrix}$$

Where,

ϕ = Mode shape

y = modal amplitude

u = total displacement

## 3.5 Free Vibration response of undamped system: Modal superposition method

$\{u_1\} = [\phi_1]\{y_1\}$,

$\{u_2\} = [\phi_2]\{y_2\}$, and

$\{u_3\} = [\phi_3]\{y_3\}$,

Total displacement,

$\{u\} = \{u_1\} + \{u_2\} + \{u_3\} + \{u_4\} + \{u_5\} + \ldots\ldots\ldots\ldots \{u_n\}$

$= [\phi_1]\, y_1 + [\phi_2]\, y_2 + [\phi_3]\, y_3 + \ldots\ldots\ldots\ldots\ldots + [\phi_n]\, y_n$

{Y} is column matrix $\begin{bmatrix} Y_1 \\ Y_2 \\ \ldots \\ Y_n \end{bmatrix}$ $and\ [\phi] is\ row\ matrix\ [\phi 1\ \phi 2\ \ldots\ldots \phi_n]$

$\{u\} = \sum_{n=1}^{N} \{\phi_n\}.Y_n$ …. …. (54)

$\{u\} = [\phi]\{Y\}$ … … … (55)

Here, Y is normal coordinate.



## 3.5 Free Vibration response of undamped system: Modal superposition method

Pre-multiplying equation (55) by $\{\phi_n\}^T[m], we\ get,$

$\{\phi_n\}^T[m]\{u\} = \{\phi_n\}^T[m][\phi]\{Y\}$

$= \{\phi_n\}^T[m][\phi]Y_1 + \{\phi_n\}^T[m][\phi]Y_2 + \ldots. + \{\phi_n\}^T[m][\phi]Y_n \ldots\ldots\ldots\ldots + \{\phi_n\}^T[m][\phi]Y_N$

Using first orthogonality condition,

$\{\phi_n\}^T[m]\{\phi_m\} = 0$, m and n being different modes

So all except one term in above equation are zero.

$\{\phi_n\}^T[m]\{u\} = \{\phi_n\}^T[m][\phi_n].Y_n$ …. …. (56)

$Y_n = \frac{\{\phi_n\}^T[m]\{u\}}{\{\phi_n\}^T[m][\phi_n]}$ … … … (57)

If vector u is time dependent then, Yn coordinates will also time dependent.

This formula cannot be directly used to determine $Y_n$ from given {u}. However, it can be used to find initial condition $Y_0$.

Completed

Undamped system!-

$$[m]\{\ddot{u}\}+[k]\{u\}=\{P(t)\} \quad \text{---(58)}$$

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}\begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \\ \ddot{u}_3 \end{Bmatrix}+\begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix}=\begin{Bmatrix} P_1(t) \\ P_2(t) \\ P_3(t) \end{Bmatrix}$$

eq

$$\left.\begin{aligned} m_1\ddot{u}_1+k_{11}u_1+k_{12}u_2+k_{13}u_3 &= P_1(t) \\ m_2\ddot{u}_2+k_{21}u_1+k_{22}u_2+k_{23}u_3 &= P_2(t) \\ m_3\ddot{u}_3+k_{31}u_1+k_{32}u_2+k_{33}u_3 &= P_3(t) \end{aligned}\right\} \text{ Coupled equations}$$





Coupled Equation!- If any one of the equation cannot be solved independently, i.e it should be solved simultaneously.

The above equations are coupled equations.

we have,

$$\{u\}=[\phi]\{Y\}$$

$$\{\ddot{u}\}=[\phi]\{\ddot{Y}\}$$

$$\therefore [m][\phi]\{\ddot{Y}\}+[k][\phi]\{Y\}=\{P(t)\} \quad \text{---(59)}$$

(27)

Pre multiplying by $\{\phi_n\}^T$ we get,

$$\{\phi_n\}^T[m][\phi]\{\ddot{Y}\} + \{\phi_n\}^T[k][\phi]\{Y_n\} = \{\phi_n\}^T\{P(t)\} \quad \text{---(60)}$$

using orthogonality conditions

$$\{\phi_n\}^T[m]\{\phi_n\}\{\ddot{Y}_n\} + \{\phi_n\}^T[k]\{\phi_n\}\{Y_n\} = \{\phi_n\}^T\{P(t)\} \quad \text{---(61)}$$

i.e. $M_n \ddot{Y}_n + K_n Y_n = P_n(t)$ ---(62)

which is SDOF equation for mode n.

where, $M_n = \{\phi_n\}^T[m]\{\phi_n\}$ — normal coordinate generalized mass

$K_n = \{\phi_n\}^T[k]\{\phi_n\}$ — " " stiffness

$P_n = \{\phi_n\}^T\{P(t)\}$ — " " load.





for mode 1,

$$M_1\ddot{Y}_1 + K_1 Y_1 = P_1(t)$$

The solution of equation of mode '1' does not depend on other modes, so it is called un-coupled equation of motion.

we have,

$$[K]\{\phi_n\} = \omega_n^2[m]\{\phi_n\}$$

Pre multiplying by $\{\phi_n\}^T$

$$\{\phi_n\}^T[k]\{\phi_n\} = \omega_n^2\{\phi_n\}[m]\{\phi_n\} \quad \text{---(63)}$$

$$\therefore K_n = \omega_n^2 M_n \quad \text{---(64)}$$

This procedure is called the mode-superposition method, or more precisely the mode displacement superposition.

## 3.7 Forced Vibration response of damped system

**Damped System:**

$$[m]\{\ddot{u}\} + [c]\{\dot{u}\} + [k]\{u\} = \{P(t)\}$$

Solving in the similar manner, we get

$$M_n \ddot{Y}_n + C_n \dot{Y}_n + K_n Y_n = P_n(t) \quad \text{---(65)}$$

which is SDOF equation of motion for damped system for mode-'n'. (uncoupled equation).

where, $C_n = \{\phi_n\}^T [c] \{\phi_n\}$

Dividing by $M_n$, we get

$$\ddot{Y}_n + \frac{C_n}{M_n}\dot{Y}_n + \frac{K_n}{M_n} Y_n = \frac{P_n(t)}{M_n}$$

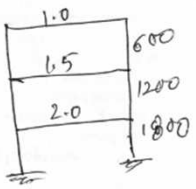
$$\text{or,} \quad \ddot{Y}_n + 2\xi_n \omega_n \dot{Y}_n + \omega_n^2 Y_n = \frac{P_n(t)}{M_n} \quad \text{---(66)}$$

where, $\xi_n = \frac{C_n}{2\omega_n M_n}$



## 3.7 Forced Vibration response of undamped system

Ex. (3)




Initial conditions

$$\{u\}_{(t=0)} = \begin{Bmatrix} 0.5 \\ 0.4 \\ 0.3 \end{Bmatrix} \text{in} \quad \& \quad \{\dot{u}\}_{(t=0)} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} \text{in/sec}$$

$$M_n = \{\phi_n\}^T [m] \{\phi_n\} \quad \& \quad [\phi] = \begin{bmatrix} 1.000 & 1.000 & 1.000 \\ 0.6486 & -0.6066 & -2.5405 \\ 0.3018 & -0.6790 & 2.4382 \end{bmatrix}$$

$$\therefore M_n = \begin{bmatrix} 1.8131 & 0 & 0 \\ 0 & 2.4740 & 0 \\ 0 & 0 & 22.596 \end{bmatrix} \quad \Big| \quad [m] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.50 & \\ 0 & 0 & 2.0 \end{bmatrix}$$

$$Y_n = \frac{\{\phi_n\}^T [m] \{u\}}{\{\phi_n\}^T [m] \{\phi_n\}} = \frac{\{\phi_n\}^T [m] \{u\}}{M_n} = M_n^{-1} \{\phi_n\}^T [m] \{u\}$$

$$M_n^T\{\phi\}^T[m] = \begin{bmatrix} 0.5515 & 0.5365 & 0.3330 \\ 0.4042 & -0.3678 & -0.5489 \\ 0.0443 & -0.1682 & 0.2159 \end{bmatrix} \quad (29)$$

$$\therefore Y_{(t=0)} = M_n^T\{\phi\}^T[m]\{u\}_{t=0} = \begin{Bmatrix} 0.5903 \\ -0.1097 \\ 0.0194 \end{Bmatrix} \text{ in}$$

$$\dot{Y}_{(t=0)} = M_n^T\{\phi\}^T[m]\{\dot{u}\}_{(t=0)} = \begin{Bmatrix} 4.829 \\ -3.310 \\ -1.519 \end{Bmatrix} \text{ in/sec}$$

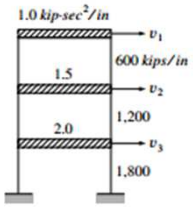
$$Y_n = \frac{\dot{Y}_{n(t=0)}}{\omega_n}\sin\omega_n t + Y_{n(t=0)}\cos\omega_n t \qquad \left[ Y_{n(t)} = Y_{n(0)}\cdot\cos\omega_n t + \frac{\dot{Y}_{n(0)}}{\omega_n}\cdot\sin\omega_n t \right]$$

$$\therefore \begin{Bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{Bmatrix} = \begin{Bmatrix} 0.3325\sin\omega_1 t \\ -0.1066\sin\omega_2 t \\ -0.0329\sin\omega_3 t \end{Bmatrix} + \begin{Bmatrix} 0.5903\cos\omega_1 t \\ -0.1097\cos\omega_2 t \\ 0.0194\cos\omega_3 t \end{Bmatrix}$$





Ex.3 Evaluate response of the given structure using mode superposition method to a sine-pulse blast-pressure load. The load may be expressed as short period impulse.




$$\begin{Bmatrix} p_1(t) \\ p_2(t) \\ p_3(t) \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \\ 2 \end{Bmatrix} (500\ kips)\cos\frac{\pi}{t_1}t \qquad \text{where} \quad t_1 = 0.02\ sec, \quad -\frac{t_1}{2} < t < \frac{t_1}{2}$$

Solution:

With this short-duration loading, it may be assumed that the response in each mode is a free vibration with its amplitude defined by the sine-pulse displacement-response spectrum. Thus during the early response era, when the effect of damping may be neglected, the modal response may be expressed as

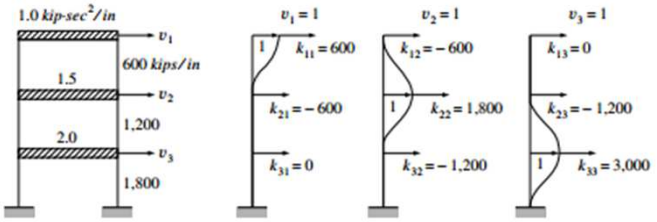
## 3.7 Forced Vibration response of undamped system

$$Y_n(t) = D_n \frac{P_{0n}}{K_n} \sin \omega_n t$$

in which

$$K_n = M_n \omega_n^2 \qquad P_{0n} = \boldsymbol{\phi}_n^T \begin{Bmatrix} 1 \\ 2 \\ 2 \end{Bmatrix} 500\ kips$$

Remind Ex.1




$$\{\omega\} = \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} = \begin{Bmatrix} 14.522 \\ 31.048 \\ 46.100 \end{Bmatrix} rad/sec \qquad [m] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} kip\cdot sec^2/in$$



## 3.7 Forced Vibration response of undamped system

$$\begin{Bmatrix} K_1 \\ K_2 \\ K_3 \end{Bmatrix} = \begin{bmatrix} 1.8131\,\omega_1^2 \\ 2.4740\,\omega_2^2 \\ 22.596\,\omega_3^2 \end{bmatrix} = \begin{Bmatrix} 382.36 \\ 2,384.9 \\ 48,019 \end{Bmatrix} kips/in \qquad \text{(b)}$$

$$\begin{Bmatrix} P_1 \\ P_2 \\ P_3 \end{Bmatrix} = \boldsymbol{\Phi}^T \begin{Bmatrix} 500 \\ 1,000 \\ 1,000 \end{Bmatrix} = \begin{Bmatrix} 1,450.40 \\ -785.59 \\ 397.80 \end{Bmatrix} kips \qquad \text{(c)}$$

Also, the impulse length-period ratios for the modes of this structure are

$$\begin{Bmatrix} \frac{t_1}{T_1} \\ \frac{t_1}{T_2} \\ \frac{t_1}{T_3} \end{Bmatrix} = \frac{0.02}{2\pi} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} = \begin{Bmatrix} 0.046 \\ 0.099 \\ 0.147 \end{Bmatrix}$$

and from Fig. 5-6, these give the following modal dynamic magnification factors:

$$\begin{Bmatrix} D_1 \\ D_2 \\ D_3 \end{Bmatrix} = \begin{Bmatrix} 0.1865 \\ 0.4114 \\ 0.6423 \end{Bmatrix} \qquad \text{(d)}$$

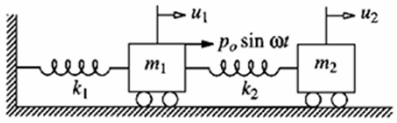
Hence, using the results given in Eqs. (b) to (d) in Eq. (a) leads to

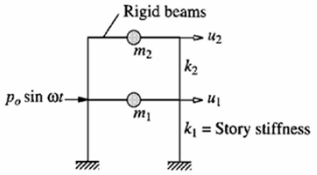
$$\begin{Bmatrix} Y_1(t) \\ Y_2(t) \\ Y_3(t) \end{Bmatrix} = \begin{Bmatrix} 0.7074 \sin \omega_1 t \\ -0.1355 \sin \omega_2 t \\ 0.0053 \sin \omega_3 t \end{Bmatrix} in \qquad \text{(e)}$$



## Additional question for exercise

Q. 1 By modal analysis determine the steady-state response of the system given below. Take m1 = 2m; m2 = m; k1 = 2k; k2 = k.



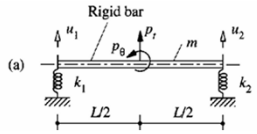

Q. 2 Consider the system of Ex.1. Determine the spring force for system of figure “a” or storey shears in the system of figure “b” without introducing equivalent static forces. Consider only steady state .

Q. 3 Repeat the Ex. 2 using equivalent static forces.

Q.4 Consider the system and excitation of Ex. 1 with modal damping ratio $\xi$. Determine the steady-state displacement amplitudes of the system.

Q.5 The dynamic response of the system as shown to the excitation given below is to be required. Determine a) displacements of nodes; b) bending moments and shears at sections a,b,c,d as function of time; c) the shraring force and bending moment diagrams at t = 0.18 sec. Take E = 29,000 ksi; I = 100 in4; L=120in; mL = 0.1672 kip-sec2/in; and P0 = 5 kips. Neglect damping. 442AKC

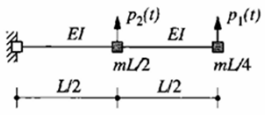
Q.6 A uniform rod of bar of total mass "m" is supported to two spring support having stiffness k1 & k2 at the two ends and subjected to dynamic forces as shown. Formulate the equation of motion with respect to displacements u1 and u2 at two ends.

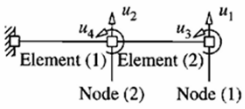





Q.7 Repeat the question no. 6 with displacement coordinates at center of beam 1st having vertical translation and 2nd rotation anticlockwise.

Q.8 A mass less cantilever beam of length L supports two lumped mass m1 & m2 at the mid point and free end. The flexural rigidity of the uniform beam is EI. With the 4 DOFs and applied forces formulate equations of motions of the system. Neglect axial and shear deformations.



## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

Response Spectra:

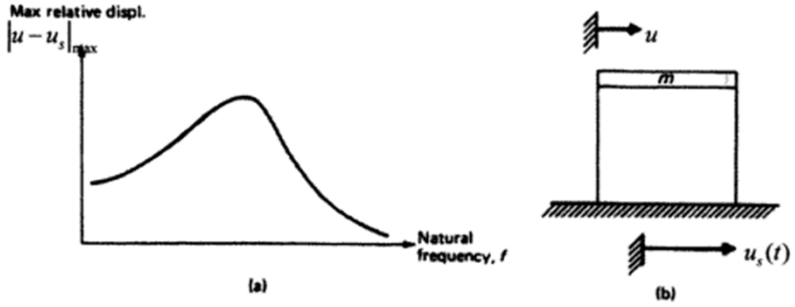



**Fig. 5.1** (a) Typical response spectrum. (b) Single-degree-of-freedom system subject to ground excitation.



## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

Response Spectra:

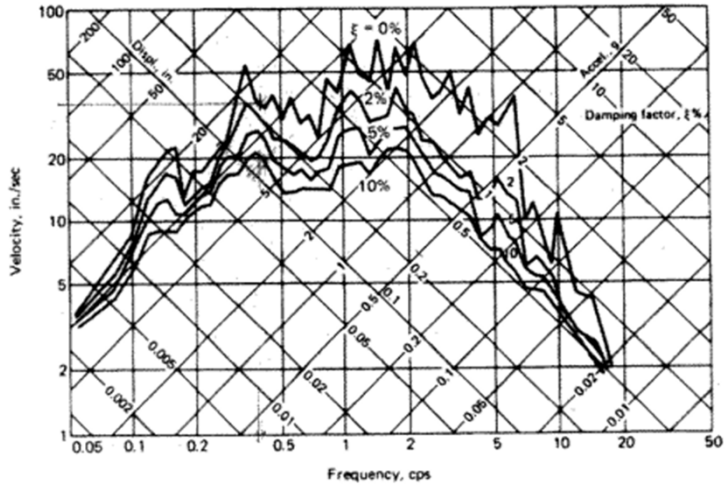



**Fig. 5.8** Response spectra for elastic system for the 1940 El Centro earthquake (from Blume et al. 1961.)

## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

Response Spectra:

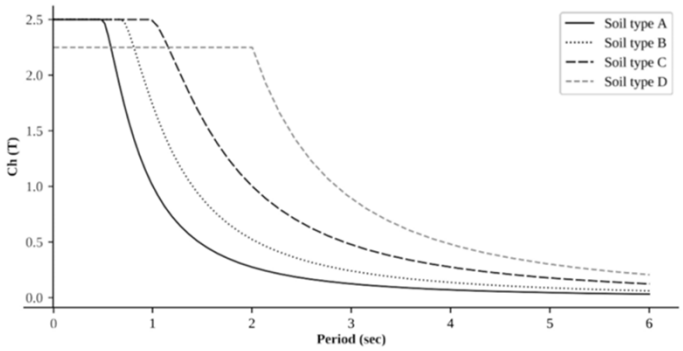



*Figure 4-1Spectral Shape Factor, $C_h(T)$ for Equivalent Static Method*



## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

Response Spectra:

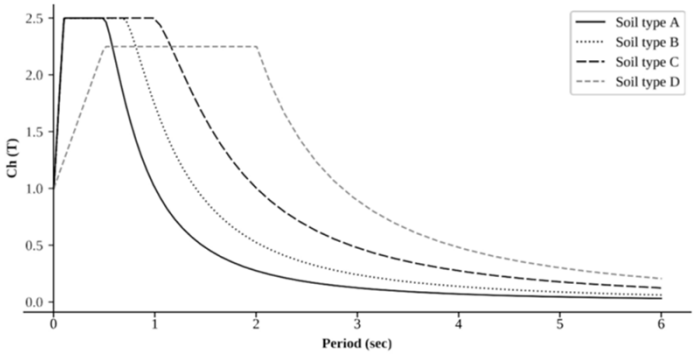



*Figure 4-2* Spectral Shape Factor, $C_h(T)$ for Modal Response Spectrum Method, Nonlinear Time History Analysis , Vertical Loading and Parts and Components

## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

The Elastic site spectra for horizontal loading shall be as given by equation 4.1(1).

$C(T) = C_h(T)\, Z\, I$ .....4.1(1)

Where,

$C_h(T)$ = Spectral Shape factor as per 4.1.2

Z = Seismic Zoning factor as per 4.1.4

I = Importance factor as per 4.1.5

**C(T)** Elastic Site Spectra for horizontal loading



## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis

For the ultimate limit state, the horizontal base shear coefficient (design coefficient), $C_d(T_1)$, shall be given by:

$$C_d(T_1) = \frac{C(T_1)}{R\mu \times \Omega u} \quad \text{.....6.1(1)}$$

Where,

$C(T_1)$ = Elastic Site Spectra as per 4.1.1

$R_\mu$ = Ductility Factor as per 5.3

$\Omega u$ = Over strength Factor for ULS as per 5.4

### 6.2 HORIZONTAL SEISMIC BASE SHEAR

The horizontal seismic base shear, V, acting at the base of the structure, in the direction being considered, shall be calculated as:

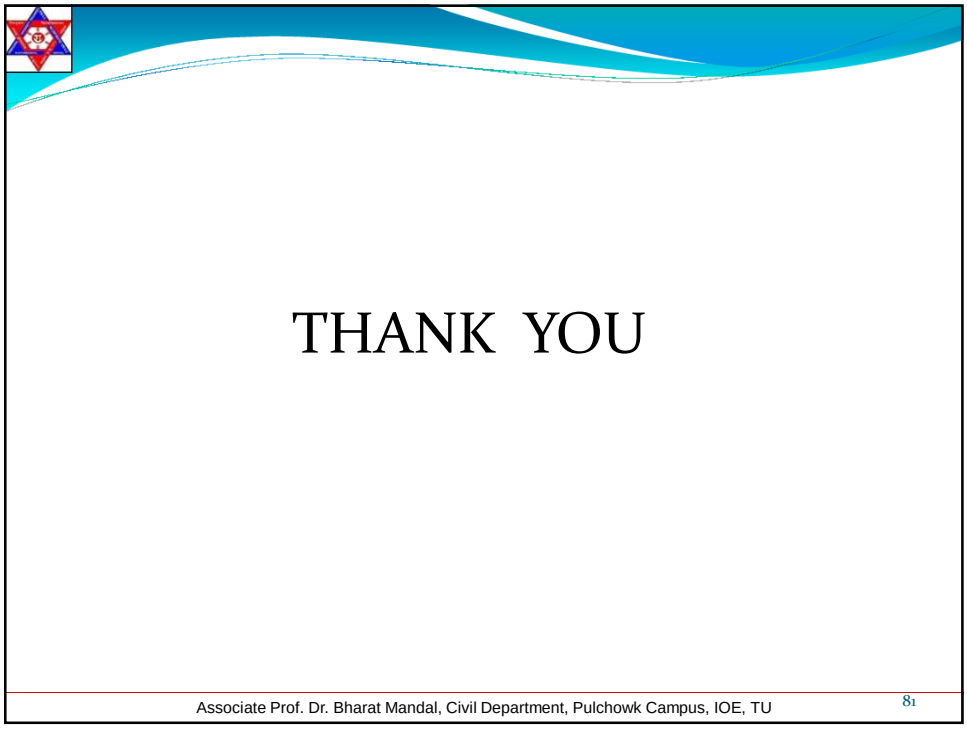
$V = C_d(T_1)\, W$ ..... 6.2(1)

### 6.3 VERTICAL DISTRIBUTION OF SEISMIC FORCES

The lateral seismic force ($F_i$) induced at each level 'i' shall be calculated as:

$$F_i = \frac{W_i h_i^k}{\sum_i^n W_i h_i^k} \times V \quad \text{.....6.3(1)}$$

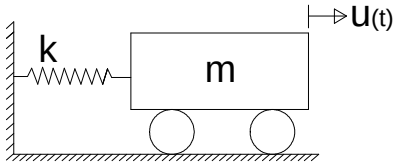
## 3.8 Dynamic analysis of linear MDOF system: Response spectra analysis


THANK YOU
Associate Prof. Dr. Bharat Mandal, Civil Department, Pulchowk Campus, IOE, TU
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## 3.9 Evaluation of natural frequencies and mode shapes (Iterative methods)
### Rayleigh's method

Rayleigh's Method:

The vibration frequency or period of a SDOF system has a controlling influence on its dynamic behaviour. Rayleigh developed a simple method of evaluating the vibration frequency for SDOF system. Basic concept is the principle of conservation of energy. The energy in a freely vibrating system must remain constant if no damping forces act to absorb it. Let us consider the free vibration motion of a undamped spring-mass system.




$u(t) = \rho \, Sin\omega_n t$, and

$\dot{u}(t) = \rho\omega_n \, Cos\omega_n t$



## 3.9 Evaluation of natural frequencies and mode shapes (Iterative methods)
### Rayleigh's method

The potential energy of this system,

$$E_p = \frac{1}{2}ku(t)^2 = \frac{1}{2}k.\rho^2 Sin^2\omega_n t \quad \ldots \quad \ldots \quad (1)$$

The Kinetic energy of this system,

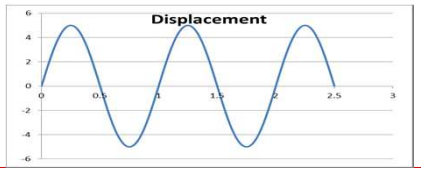
$$E_k = \frac{1}{2}m\dot{u}(t)^2 = \frac{1}{2}m.\rho^2\omega_n^2.Cos^2\,\omega_n t \quad \ldots \quad \ldots \quad (2)$$

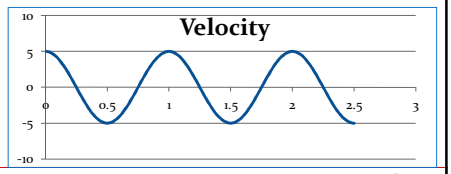
At the time when t = T/4 = $\pi/(2\omega_n)$, the kinetic energy is zero and the potential energy reaches its maximum value.

$$E_{pmax} = \frac{1}{2}k.\rho^2 \quad \ldots \quad \ldots \quad (3)$$

Similarly, at the time when t = T/2 = $\pi/\omega_n$, the potential energy vanishes and the kinetic energy becomes maximum.

$$E_{kmax} = \frac{1}{2}m.\rho^2\omega_n^2 \quad \ldots \quad \ldots \quad (4)$$



## 3.9 Evaluation of natural frequencies and mode shapes (Iterative methods)
### Rayleigh's method

Therefore, if the total energy in the vibrating system remains constant as it must in undamped free vibration, it is then apparent that the maximum potential energy must be equal to the maximum kinetic energy, $E_{Pmax} = E_{Kmax}$. Then,

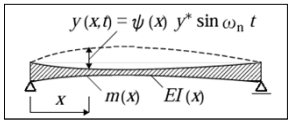
$$\omega_n^2 = \frac{k}{m} \quad \text{or,} \quad \omega_n = \sqrt{k/m} \ldots \quad \ldots \quad (5)$$

This is of course, the same frequency equation.

Approximate analysis of General Systems

Consider a non-uniform simple beam having an infinite numbers of degree of freedom. Noting the harmonic vibration of the generalized coordinate in free vibration, the displacement of the beam can be expressed by

$x(x,t) = \psi_{(x)}.Y.\sin\omega_n t$




Where, $\psi_{(x)}$ – is the shape function

Y – is the generalized coordinate



## 3.9 Evaluation of natural frequencies and mode shapes (Iterative methods)
### Rayleigh's method

This equation express the assumption that shape of the vibrating beam does not change with time only the amplitude of motion varies, and it varies harmonically in free vibration. The assumption of the shape function $\psi_{(x)}$ effectively reduces the beam to a SDOF system.

Note that 1st mode of vibration shape is either




The strain energy of this flexural system is given by,

$$E_{p(t)} = \frac{1}{2}\int EI_{(x)}\{\frac{\partial^2 u(x,t)}{\partial x^2}\}^2 dx$$

Potential energy = strain energy would be max., when displacement is max. i.e., $Y.\sin\omega_n t = Y_0$,

Therefore, $E_{p(t)} = \frac{1}{2}\int EI_{(x)}\{\frac{\partial^2(\psi_{(x)}.Y_0)}{\partial x^2}\}^2 dx = \frac{1}{2}\int EI_{(x)}\{\psi_{(x)}.Y_0\}^2 dx$

$$Or, \quad E_{p(t)} = \frac{1}{2}Y_0^2 \int_0^L EI_x(\psi''_{(x)})^2 dx \quad \ldots. \quad \ldots. \quad (6)$$

Kinetic energy is maximum, when velocity is maximum,

$\dot{u}(x,t) = \psi_x . Y_0 . \omega . Cos\omega t, \quad and \quad \dot{u}(x,t)_{max} = \psi_x . Y_0 . \omega$

$$E_{k(t)} = \frac{1}{2}\int m_{(x)}\{\dot{u}(x,t)\}^2 dx = \frac{1}{2}\int m_{(x)}\{\psi_x . Y_0 . \omega\}^2 dx$$

$$= \frac{1}{2}Y_0^2\omega^2 \int m_{(x)}[\psi_{(x)}]^2 dx \quad .... \quad ... \quad (7)$$

Equating equation (6) and (7), we get,

$$\omega^2 = \frac{\int EI_{(x)}\{\psi''(x)\}^2 dx}{\int m_{(x)}\{\psi_{(x)}\}^2 dx} \quad .... \quad ..... \quad (8)$$

Comparing this equation with $\omega^2 = \frac{k}{m}$, represents

$$\int \boldsymbol{EI_{(x)}\{\psi''(x)\}^2 dx} = \boldsymbol{k^*, Generalized\ stiffness}$$

$$\int \boldsymbol{m_{(x)}\{\psi_{(x)}\}^2 dx} = \boldsymbol{m^*, Generalized\ mass}$$

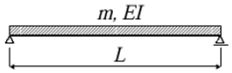




Thus, Rayleigh's method can be directly applied to any SDOF system of the generalized form and the frequency can be given by

$$\omega^2 = \frac{k^*}{m^*} \quad ... \quad ... \quad (9)$$

Ex. 2.5 Derive the natural frequency of a simply supported uniform beam, using Rayleigh's method.




Assume first the shape function as

$$\psi_{(x)} = x^2 - Lx, \quad Therefore, \quad \psi''_{(x)} = 2$$

Then,

$$k^* = \int_0^L EI_{(x)}[\psi''_{(x)}]^2 dx = \int_0^L EI.(2)^2 dx = 4EI[x]_0^L = 4EIL$$

$$m^* = \int_0^L m_{(x)}[\psi_{(x)}]^2 dx = m\int_0^L (x^2 - Lx)dx = m\int_0^L (x^4 - 2Lx^3 + L^2x^2)$$

$$m^* = m\left[\frac{x^5}{5} - 2L\frac{x^4}{4} + L^2\frac{x^3}{3}\right] = \frac{mL^5}{30}$$

$$\omega = \frac{k^*}{m^*} = \frac{4EIL}{mL^5/30} = 120\frac{EI}{mL^4}$$

This is not good estimation. Because curvature of the beam is uniform along the length. i.e. $\psi''_{(x)} = 2$.

Again assume shape function,

$$\psi_{(x)} = x^4 - 2Lx^3 + L^3x$$
$$\psi'_{(x)} = 4x^3 - 6Lx^2 + L^3$$
$$\psi''_{(x)} = 12x^2 - 12Lx^1$$

Then, $k^* = \int_0^L EI_{(x)}\left[\psi''_{(x)}\right]^2 dx = \frac{24}{5}EIL^5$



$$m^* = \int_0^L m_{(x)}\left[\psi_{(x)}\right]^2 dx = \int_0^L m\left(x^4 - 2Lx^3 + L^3x\right)^2 dx \quad (13)$$

$$= m\left[\frac{x^5}{5} - 2L\frac{x^4}{4} + L^3\frac{x^2}{2}\right]_0^L$$

$$= m\cdot L^5\left(\frac{1}{5} - \frac{1}{2} + \frac{1}{2}\right)$$

$$= m\int_0^L \left(x^8 + 4L^2x^6 + L^6x^2 - 4Lx^7 - 4L^4x^4 + 2L^3x^5\right) dx$$

$$= m\left[\frac{x^9}{9} + 4L^2\frac{x^7}{7} + L^6\frac{x^3}{3} - 4L\frac{x^8}{8} - 4L^4\frac{x^5}{5} + 2L^3\frac{x^6}{6}\right]_0^L$$

$$= mL^9\left(\frac{1}{9} + \frac{4}{7} + \frac{1}{3} - \frac{1}{2} - \frac{4}{5} + \frac{1}{3}\right) = \frac{31}{630}mL^9$$

$$\therefore \omega^2 = \frac{k^*}{m^*} = \frac{3024}{31}\frac{EI}{mL^4} \approx 97.5\frac{EI}{mL^4}$$

Let us assume shape function (3)

$$\psi(x) = \sin\frac{\pi x}{L}, \quad \therefore \psi'(x) = \frac{\pi}{L}\cos\frac{\pi x}{L}, \quad \psi''(x) = -\frac{\pi^2}{L^2}\sin\frac{\pi x}{L}$$

$$\therefore k^* = \int_0^L EI(x)[\psi''(x)]^2 dx = EI\int_0^L \left(-\frac{\pi^2}{L^2}\sin\frac{\pi x}{L}\right)^2 dx$$

$$= EI\frac{\pi^4}{L^4}\int_0^L \frac{1}{2}\left(1-\cos\frac{2\pi x}{L}\right)dx$$

$$= EI\frac{\pi^4}{2L^4}\left[x - \frac{L}{2\pi}\sin\frac{2\pi x}{L}\right]_0^L = EI\frac{\pi^4}{2L^4}\left(L - \frac{L}{2\pi}\sin 2\pi\right)$$

$$= \frac{\pi^4}{2}\cdot\frac{EI}{L^3}$$

$$m^* = \int_0^L m(x)[\psi(x)]^2 dx = m\int_0^L \left(\sin\frac{\pi x}{L}\right)^2 dx = \frac{m}{2}\int_0^L \left(1-\cos\frac{2\pi x}{L}\right)dx$$

$$= \frac{m}{2}\left[x - \frac{L}{2\pi}\sin\frac{2\pi x}{L}\right]_0^L = \frac{mL}{2}$$

$$\therefore \omega^2 = \frac{k^*}{m^*} = \frac{\pi^4}{2}\cdot\frac{EI}{L^3}\cdot\frac{2}{mL} \approx 9.7\frac{EI}{mL^4}$$





## Selection of shape function:

Shape function have to satisfies the geometric boundary conditions along with equilibrium.

The true vibration shape will yield the lowest frequency.

The correct vibration shape $\psi_{(x)}$ is the deflected shape that results from a loading $P_{(x)}$ proportional to m(x) $\psi_{(x)}$.

Improved Rayleigh Method:

Method R00:
Let us assume the arbitrary shape function which satisfy the geometric boundary conditions.

$$u^0(x,t) = \psi_{(x)}^{(0)}.Y_0^{(0)}.Sin\omega_n t \quad \ldots \quad \ldots \quad (10)$$

The potential and kinetic energy, are given by,

$$E_{Pmax} = \frac{1}{2}\int EI_{(x)}\left\{\frac{\partial^2 u^{(0)}}{\partial x^2}\right\}^2 dx = \frac{Y_0^{(0)^2}}{2}\int EI_{(x)}\{\psi^{"(0)}\}^2 dx \ldots(11)$$

$$E_{Kmax} = \frac{1}{2}\int m_{(x)}\{\mathring{u}^{(0)}\}^2 dx = \frac{Y_0^{(0)^2}}{2}\omega^2\int m_{(x)}\{\psi^{(0)}\}^2 dx \ldots(12)$$

$$\omega^2 = \frac{\int EI_{(x)}\{\psi^{"(0)}\}^2 dx}{\int m_{(x)}\{\psi^{(0)}\}^2 dx} \quad \ldots \quad \ldots \quad \ldots \quad (13)$$



The inertia force at the time of max displacement is,

$$P_{(x)}^{(0)} = \omega^2.m_{(x)}.u^{(0)} = Y_0^{(0)}\omega^2.m_{(x)}.\psi^{(0)} \quad \ldots \quad \ldots \quad (14)$$

The deflection caused by this loading may be expressed as,

$$u^{(1)} = \omega^2.\frac{u^{(1)}}{\omega^2} = \omega^2.\frac{Y_0^{(1)}\psi^{(1)}}{\omega^2} = \omega^2\psi^{(1)}\acute{Y}_0^{(1)} \quad \ldots \quad \ldots \quad (15)$$

The potential energy of the strain produced by this loading is given by

$$E_{Pmax} = \frac{1}{2}\int P_{(x)}^{(0)}u^{(1)}dx = \frac{1}{2}Y_0^{(0)}\acute{Y}_0^{(1)}\omega^4\int m_{(x)}\psi^{(0)}\psi^{(1)}dx \ldots..(16)$$

Equating this to the kinetic energy given by originally assumed shape leads to the improved Rayleigh frequency expression, R01,

$$\omega^2 = \frac{Y_0^{(0)}\int m_{(x)}\{\psi^{(0)}\}^2 dx}{\acute{Y}_0^{(1)}\int m_{(x)}\psi^{(0)}\psi^{(1)}dx} \quad \ldots \quad \ldots. \quad (17)$$

It gives improved accuracy.

Method R11

Kinetic energy from calculated shape $u^{(1)}$

$$E_{Kmax} = \frac{1}{2}\int m_{(x)}\{\mathring{u}^{(1)}\}^2 dx = \frac{\omega^6}{2}\{\acute{Y}_0^{(1)}\}^2 \int m_{(x)}\{\psi^{(1)}\}^2 dx \quad \ldots (18)$$

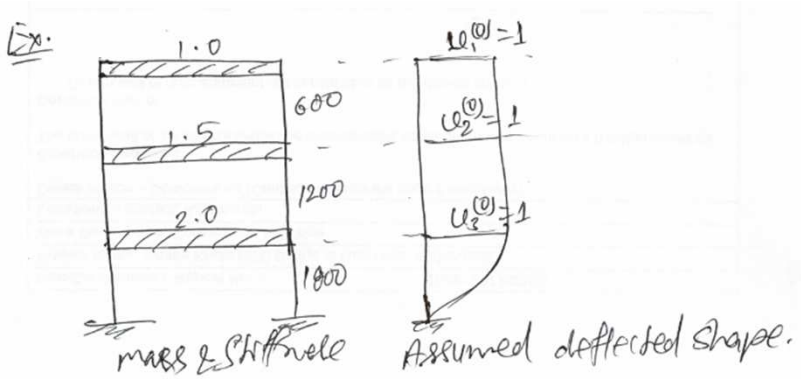
Equating equation (18) to equation (16), we get

$$\omega^2 = \frac{Y_0^{(0)}}{\acute{Y}_0^{(1)}} \frac{\int m_{(x)}\psi^{(0)}\psi^{(1)} dx}{\int m_{(x)}\{\psi^{(1)}\}^2 dx} \quad \ldots \quad \ldots . \quad (19)$$

Is better approximation.







Standard method Koo

$U_1^{(0)} = U_2^{(0)} = U_3^{(0)} = Y_0^{(0)} \cdot \psi^{(0)} = 1$

where, $\psi^{(0)} = Y_0^{(0)} = 1$ (masses are lumped)

$$E_{Kmax}^{(0)} = \frac{1}{2}\int m(x)\{\dot{U}(x)^{(0)}\}^2 dx = \frac{1}{2}\sum m_i [\dot{U}_i^{(0)}]^2$$

$$= \frac{1}{2}\omega^2 \{Y_0^{(0)}\}^2 \sum m_i [\psi_i^{(0)}]^2$$

$$= \frac{1}{2}\omega^2 \cdot (1)^2 \{1\times(1)^2 + 1.5\times(1)^2 + 2\times(1)^2\}$$

$$= 2.25\omega^2$$





$$Ep_{max}^{(0)} = \frac{1}{2}\sum k_i [\Delta U_i^{(0)}]^2$$

$$= \frac{1}{2}\{Y_0^{(0)}\}^2 \sum k_i \{\Delta\psi_i^{(0)}\}^2$$

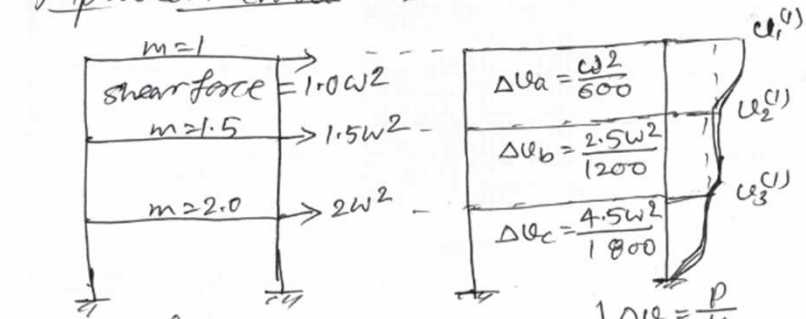
$$= \frac{1}{2}\times(1)^2[600\times 0^2 + 1200\times 0^2 + 1800\times(1)^2]$$

$$= 900$$

Equating, $E_{Kmax}^{(0)} = Ep_{max}^{(0)} \Rightarrow \omega^2 = 400$

$\therefore \omega = 20$ rad/sec.

Improved method Ro1:-




Inertia force:

$P_1^{(1)} = \omega^2 m_1 u_1^{(1)} = 1 \times 1 \times \omega^2 = 1\omega^2$

$P_2^{(1)} = \omega^2 m_2 u_2^{(1)} = 1.5 \times 1 \times \omega^2 = 1.5\omega^2$

$P_3^{(1)} = \omega^2 m_3 u_3^{(1)} = 2.0 \times 1 \times \omega^2 = 2\omega^2$

$\Delta = \frac{P}{k}$

$\therefore \Delta a = \frac{1.0 \times \omega^2}{600} = \frac{\omega^2}{600}$

$\Delta b = \frac{2.5\omega^2}{1200}$

$\Delta c = \frac{4.5\omega^2}{1800}$





The deflections:

(4

$u_3^{(1)} = \Delta c = \frac{4.5\omega^2}{1800} = \frac{9\omega^2}{3600}$

$u_2^{(1)} = \Delta b + u_3^{(1)} = \frac{2.5\omega^2}{1200} + \frac{4.5\omega^2}{1800} = \frac{16.5\omega^2}{3600}$

$u_1^{(1)} = \Delta a + u_2^{(1)} = \frac{\omega^2}{600} + \frac{16.5\omega^2}{3600} = \frac{22.5\omega^2}{3600}$

$E_{P_{max}} = \frac{1}{2} \sum P_{(i)}^{(1)} u_{(i)}^{(1)} = \frac{1}{2} \bar{X}_0^{(1)} X_0^{(1)} \omega^4 \sum m_i \psi_i^{(1)} \psi_i^{(1)}$

$= \frac{1}{2} \times 1 \times \omega^4 \times \bar{X}_0^{(1)} [1 \times 1 \times 1.5 + 1 \times 1.5 \times 0.733 + 2 \times 1 \times 0.4]$

$u_1^{(1)} = \psi_1^{(1)} X_0^{(1)} \Rightarrow \frac{22.5}{3600}\omega^2 = 1 \times X_0^{(1)} \Rightarrow X_0^{(1)} = \frac{22.5\omega^2}{3600}$

or $\bar{X}_0^{(1)} \cdot \omega^2 = X_0^{(1)} = \frac{22.5\omega^2}{3600} \Rightarrow \bar{X}_0^{(1)} = \frac{22.5}{3600}$

$u_2^{(1)} = \psi_2^{(1)} \cdot X_0^{(1)}$ or, $\frac{16.5}{3600}\omega^2 = \psi_2^{(1)} \cdot \frac{22.5\omega^2}{3600} \Rightarrow \psi_2^{(1)} = 0.733$

Similarly $\psi_3^{(1)} = 0.40$

$E_{p_{max}} = \frac{\omega^4}{2} \bar{Y}_0^{(1)} \times 2.9$

Equating $E_{p_{max}}^{(1)} = E_{k_{max}}^{(1)}$

$\frac{\omega^4}{2} \bar{Y}_0^{(1)} \times 2.9 = \frac{1}{2} \omega^2 \times 4.5 \; 2.25\omega^2 \Rightarrow \omega^2 = \frac{2.25 \times 2}{2.9} \times \frac{1}{\bar{Y}_0^{(1)}}$

$\therefore \omega^2 = \frac{4.5}{2.9} \times \frac{3600}{22.5} = 248.27$

$\therefore \omega = 15.75$ rad/sec.

Improved Method Rii:-

$E_{k_{max}}^{(1)} = \frac{\omega^6}{2} \times \{\bar{Y}_0^{(1)}\}^2 \sum m_i \{\psi_i^{(1)}\}^2$

$= \frac{\omega^6}{2} \times \left(\frac{22.5}{3600}\right)^2 [1 \times (1)^2 + 1.5 \times (0.733)^2 + 2.0 \times (0.4)^2]$

$= \frac{\omega^6}{2} \times \left(\frac{22.5}{3600}\right)^2 \times 2.1259$

$E_{k_{max}}^{(1)} = E_{p_{max}}^{(1)} \Rightarrow \frac{\omega^6}{2} \times \left(\frac{22.5}{3600}\right)^2 \times 2.1259 = \frac{\omega^4}{2} \times \left(\frac{22.5}{3600}\right) \times 2.9$

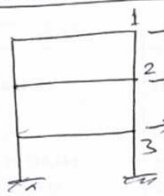
$\therefore \omega^2 = 218.26 \Rightarrow \omega = 14.77$ rad/sec.





Practical Vibration Analysis (III)

Forced Vibration:




Then, $\{P(t)\} = \begin{Bmatrix} P_1(t) \\ P_2(t) \\ P_3(t) \end{Bmatrix} = \begin{Bmatrix} 45 \text{ kN} \\ 25\sin 12t \\ 0 \end{Bmatrix}$

We have, $M_n(t) = \{\phi_n\}^T [m] \{\phi_n\}$

$P_n(t) = \{\phi_n\}^T \{P(t)\}$

$\therefore P_1(t) = \{\phi_1\}^T \{P(t)\} = \{\phi_{11} \; \phi_{21} \; \phi_{31}\} \cdot \begin{Bmatrix} 45 \text{ kN} \\ 25\sin 12t \\ 0 \end{Bmatrix}$

$= \phi_{11} \cdot 45 + \phi_{21} \cdot 25 \sin 12t$

Similarly, $P_2(t) = \{\phi_2\}^T \cdot \{P(t)\} = \{\phi_{12} \; \phi_{22} \; \phi_{32}\} \begin{Bmatrix} 45 \\ 25\sin 12t \\ 0 \end{Bmatrix}$

$= \phi_{12} \cdot 45 + \phi_{22} \cdot 25 \sin 12t$

Stodla Method:

Assumed mode shape is adjusted untill the true mode shape is obtained and then the frequency of vibration is calculated.

Let a = flexibility coefficient, then

k.a = a.k = 1

In matrix form, [a].[k] = [k].[a] = [1]

Or, [a] = $[k]^{-1}$ and [k] = $[a]^{-1}$

We have, $([k]-\omega^2[m])\{\hat{u}\} = \{0\}$

Pre-multiplying by $\frac{1}{\omega^2}[a]$ , we get,

$$\left(\frac{1}{\omega^2}[a][k]-[a][m]\right)\{\hat{u}\} = \frac{1}{\omega^2}[a].\{0\} = \{0\}$$

$$\left(\frac{1}{\omega^2}[I]-[a][m]\right)\{\hat{u}\} = \{0\}$$

$$\frac{1}{\omega^2}\{\hat{u}\} = [a][m]\{\hat{u}\} \quad \ldots. \quad (1)$$





Say Dynamic matrix, [D] = [a][m] … …. (2)

From equation (1) and (2),

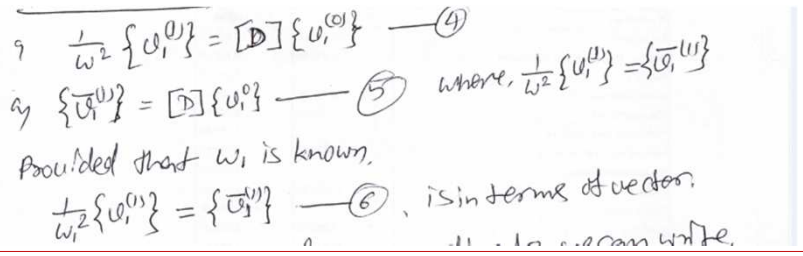
$$\frac{1}{\omega^2}\{\hat{u}\} = [D]\{\hat{u}\} \quad \ldots \quad \ldots. \quad (3)$$

$u_1^0 \Rightarrow$ 1 = indicates mode and

0 = indicates origin (trial shape at the beginning)

[Trial shape is assumed as near as possible to the first mode (as like cantilever structure). Also it must satisfy the boundary condition]

1st iteration:



$\omega_1 < \omega_2 < \omega_3$

Similarly in term of one coordinate we can write,

$$\bar{v}_{k1}^{(1)} = \frac{1}{\omega_1^2} v_{k1}^{(0)} \quad \text{---(7)} \quad \{k_1 = 1,2,3\}$$

[Eg. $k=3$, $\omega_1^2 = \frac{v_{31}^{(0)}}{\bar{v}_{31}^{(1)}}$]

or $\omega_1^2 = \frac{v_{k1}^{(0)}}{\bar{v}_{k1}^{(1)}}$ ---(8)

or $\left(\frac{v_{k1}^{(0)}}{\bar{v}_{k1}^{(1)}}\right)_{min} < \omega_1^2 < \left(\frac{v_{k1}^{(0)}}{\bar{v}_{k1}^{(1)}}\right)_{max}$ ---(9)

For a value of k, $\left(\frac{v_{k1}^{(0)}}{\bar{v}_{k1}^{(1)}}\right)$ is min.

For another value of k, $\left(\frac{v_{k1}^{(0)}}{\bar{v}_{k1}^{(1)}}\right)$ is max.

or $\omega_1^2 = \frac{\{\bar{v}_1^{(1)}\}^T [m] \{v_1^{(0)}\}}{\{\bar{v}_1^{(1)}\}^T [m] \{\bar{v}_1^{(1)}\}}$ ---(10)

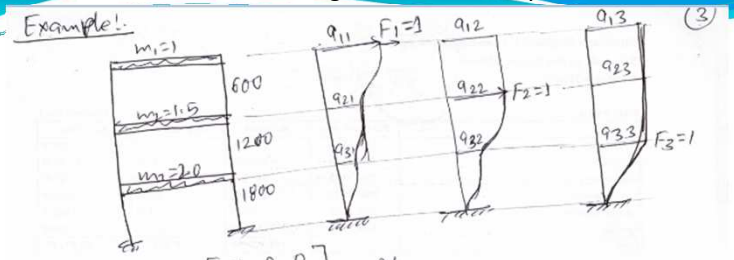
After 3 cycles, $\{\bar{v}_1^{(3)}\} = \frac{1}{\omega_1^2} \{v_1^{(3-1)}\}$ ---(11)

or $\omega_1^2 = \frac{v_{k1}^{(3-1)}}{\bar{v}_{k1}^{(3)}}$ ---(12)





Example:-




Solution:

$$[m] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} \text{ kn-s}^2/\text{in}$$

Flexibility matrix, $[a] = [k]^{-1} = \frac{1}{3600}\begin{bmatrix} 11 & 5 & 2 \\ 5 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$ in/kip

$$[D] = [a][m] = \frac{1}{3600}\begin{bmatrix} 11 & 5 & 2 \\ 5 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \frac{1}{3600}\begin{bmatrix} 11 & 7.5 & 4 \\ 5 & 7.5 & 4 \\ 2 & 3 & 4 \end{bmatrix} \text{ sec}^2$$

Assume the initial shape as $\begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}$

$$[D]\{v_1^0\} = \frac{1}{3600}\begin{bmatrix} 11 & 7.5 & 4 \\ 5 & 7.5 & 4 \\ 2 & 3 & 4 \end{bmatrix} \cdot \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix} = \frac{1}{3600}\begin{Bmatrix} 22.5 \\ 16.5 \\ 9.0 \end{Bmatrix} = \{\bar{v}_1^{(1)}\}$$

$\{\bar{v}_1^{(1)}\} = [D]\{v_1^0\}$

## Multi Degree of Freedom System

To normalize the matrix, divide by highest value.

$$\begin{Bmatrix}22.5\\16.5\\9.0\end{Bmatrix} \Rightarrow 22.5\begin{Bmatrix}1.0\\0.73\\0.40\end{Bmatrix} \quad \therefore \{\phi_1^{(1)}\} = \begin{Bmatrix}1.0\\0.73\\0.40\end{Bmatrix}$$

Again, $[D]\{\phi_1^{(1)}\} = \frac{1}{3600}\begin{bmatrix}11 & 7.5 & 4\\5 & 7.5 & 4\\2 & 3 & 4\end{bmatrix}\begin{Bmatrix}1.0\\0.73\\0.40\end{Bmatrix} = \frac{1}{3600}\begin{Bmatrix}18.075\\12.075\\5.790\end{Bmatrix} = \bar{\phi}_1^{(2)}$

Normalizing again,

$$\begin{Bmatrix}18.075\\12.075\\5.790\end{Bmatrix} \Rightarrow 18.075\begin{Bmatrix}1.00\\0.67\\0.32\end{Bmatrix} \quad \therefore \{\phi_1^{(2)}\} = \begin{Bmatrix}1.00\\0.67\\0.32\end{Bmatrix}$$



## Multi Degree of Freedom System

Again,

$$[D]\{\phi_1^{(2)}\} = \frac{1}{3600}\begin{bmatrix}11 & 7.5 & 4\\5 & 7.5 & 4\\2 & 3 & 4\end{bmatrix}\begin{Bmatrix}1.00\\0.67\\0.32\end{Bmatrix} = \frac{1}{3600}\begin{Bmatrix}17.305\\11.305\\5.29\end{Bmatrix} = \bar{\phi}_1^{(3)}$$

Normalizing again.

$$\begin{Bmatrix}17.305\\11.305\\5.29\end{Bmatrix} \Rightarrow 17.305\begin{Bmatrix}1.00\\0.65\\0.302\end{Bmatrix} \quad \therefore \{\phi_1^{(3)}\} = \begin{Bmatrix}1.00\\0.65\\0.302\end{Bmatrix}$$

Again, $[D]\{\phi_1^{(3)}\} = \frac{1}{3600}\begin{bmatrix}11 & 7.5 & 4\\5 & 7.5 & 4\\2 & 3 & 4\end{bmatrix}\begin{Bmatrix}1.00\\0.65\\0.302\end{Bmatrix} = \frac{1}{3600}\begin{Bmatrix}17.083\\11.083\\5.158\end{Bmatrix} = \bar{\phi}_1^{(4)}$

Normalizing again.

$$\begin{Bmatrix}17.083\\11.083\\5.158\end{Bmatrix} \Rightarrow 17.083\begin{Bmatrix}1.00\\0.649\\0.302\end{Bmatrix} \quad \therefore \{\phi_1^{(4)}\} = \begin{Bmatrix}1.00\\0.649\\0.302\end{Bmatrix}$$

## Multi Degree of Freedom System

Again,

$$[D]\{v_1^4\} = \frac{1}{3600}\begin{bmatrix} 11 & 7.5 & 4 \\ 5 & 7.5 & 4 \\ 2 & 3 & 4 \end{bmatrix}\begin{Bmatrix} 1.00 \\ 0.649 \\ 0.302 \end{Bmatrix} = \frac{1}{3600}\begin{Bmatrix} 17.076 \\ 11.076 \\ 5.155 \end{Bmatrix} = \vec{v}_1^{(5)}$$

Normalizing again,

$$\begin{Bmatrix} 17.076 \\ 11.076 \\ 5.155 \end{Bmatrix} \Rightarrow 17.076\begin{Bmatrix} 1.00 \\ 0.649 \\ 0.302 \end{Bmatrix}, \quad v_1^{(5)} = \begin{Bmatrix} 1.00 \\ 0.649 \\ 0.302 \end{Bmatrix}$$

After 4th iteration,

$$\omega_1^2 = \frac{v_{11}^{(1)}}{\bar{v}_{11}^{(5)}} = \frac{1}{17.076 \times \frac{1}{3600}} = \frac{3600}{17.076} = 210.82$$

$\therefore \omega_1 = 14.52$ rad/sec.

$$(\omega_1^2)_{min} = \frac{1}{22.5} \times 3600 = 160 \quad \left[\frac{v_{11}^{0}}{\bar{v}_{11}^{(1)}}\right]$$

$$(\omega_1^2)_{max} = \frac{v_{31}^{(0)}}{\bar{v}_{31}^{(1)}} = \frac{1}{9} \times 3600 = 400$$

$\therefore 160 < \omega_1^2 < 400$

$\therefore 12.65 < \omega_1 < 20$



## Multi Degree of Freedom System

From eqn (10)

$$\omega_1^2 = \frac{\{\bar{v}_1^{(1)}\}^T [m] \{v_1^{(0)}\}}{\{\bar{v}_1^{(1)}\}^T [m] \{\bar{v}_1^{(1)}\}}$$

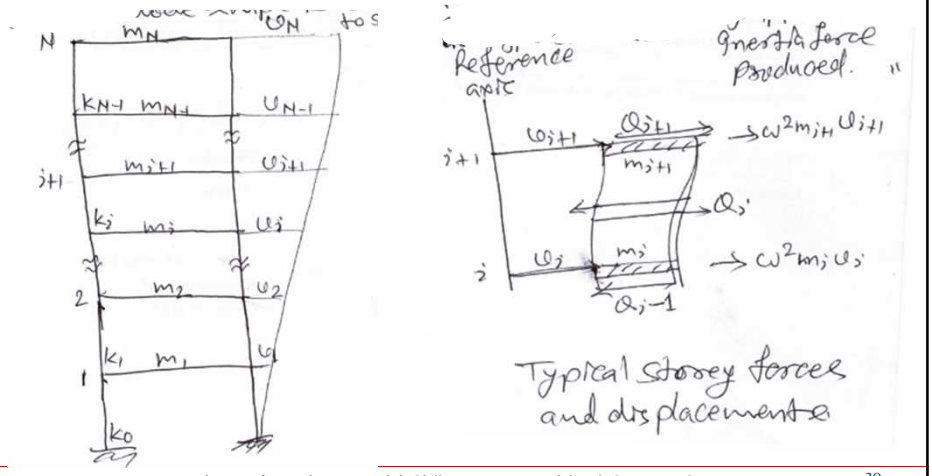
$$= \frac{\{22.5 \quad 16.5 \quad 9.0\}\begin{bmatrix} 1.00 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix}\begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}}{\{22.5 \quad 16.5 \quad 9.0\}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 2 \end{bmatrix}\begin{Bmatrix} 22.5 \\ 16.5 \\ 9.0 \end{Bmatrix} \times \frac{1}{3600}} = \frac{65.25}{1076.625} \times 3600 = 218.182$$

$\therefore \omega_1 = 14.77$ rad/sec.

Holzer Method: (Iterative method)

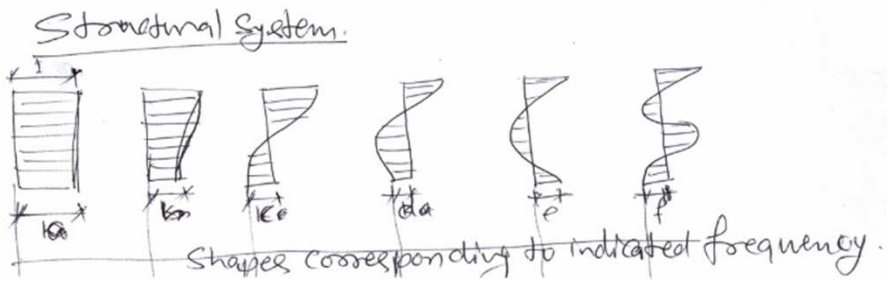
The frequency is adjusted so that the true frequency is established and the mode shape is evaluated simultaneously.

Only applicable to shear type structure.

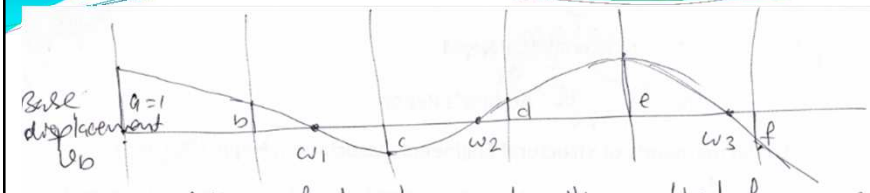






(6)




Variation of displacement with applied frequency (maintaining unit displacement at top)

Deflection ($\Delta v_N$) → change in absolute displacement between two floor.

[As we know, from the initial condition, base displacement is zero. So, we assume a certain value at the begining '1' and carry on iteration till base displacement is zero. The corresponding frequency is the actual frequency ($\omega_1, \omega_2, \omega_3$)]



Procedure:

(i) Assume $\omega_j$ (as far as possible, minimum)

(ii) The inertia force $f_I$ produced by the n-th mass is equal to the shear force $Q_n$ just below the n-th mass, and is given by,

$$f_{In} = Q_n = \omega_j^2 M_n v_n$$

(iii) The deflection caused by the shear force $Q_n$ is

$$\Delta v_n = Q_n / K_n = \frac{\omega_j^2 M_n v_n}{K_n} = v_n - v_{n-1}$$

(iv) Then, the (n-1)th element of the mode shape vector is

$$v_{n-1} = v_n - \Delta v_n$$

(v) The shear force just below the (n-1)th mass is

$$Q_{n-1} = f_{In} + f_{I(n-1)} = \sum_{i=n-1}^{n} \omega^2 M_i v_i$$

(vi) Then the deflection $\Delta v_{n-1}$ caused by the shear force $Q_{n-1}$ : $\Delta v_{n-1} = \frac{Q_{n-1}}{K_{n-1}} = v_{n-1} - v_{n-2}$

(vii) The $(n-2)^{th}$ element of the mode shape vector is (7)

$$\phi_{n-2} = \phi_{n-1} - \Delta\phi_{n-1}$$

(viii) Repeat the step (v) to (vii), we get

$$\Delta\phi_1 = \frac{Q_1}{k_1} = \sum_{j=1}^{n} \omega^2 m_j \phi_j / k_i$$

(ix) $\phi_0 = \phi_1 - \Delta\phi_1$

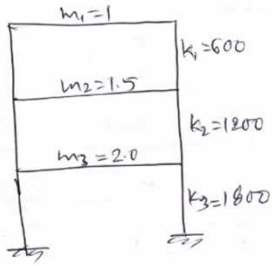
If $u_o = 0$, the initial assumption of $\omega$ is the correct frequency of vibration.

If this value is positive ($u_o > 0$), the assumed $\omega$ is smaller than the true $\omega$ in the case of odd modes ($\omega_1$, $\omega_3$, ....) or the assumed $\omega$ is larger than the true $\omega$ in the case of even modes ($\omega_2$, $\omega_4$, ....)





Ex.




Trial – I

Assume $\omega^2 = 100$

| $\phi$ | $\Delta\phi$ | Q | $f_I$ |
|---|---|---|---|
| 1.00 | | | 100 |
| | 0.167 | 100 | |
| 0.833 | | 224.95 | 124.95 |
| | 0.187 | 224.95 | |
| 0.646 | | | 129.2 |
| | 0.197 | 354.15 | |
| 0.449 | | | |

Assume $\phi_n = 1$, Calculate $f_{I_n}$, $Q_n$ and $\Delta\phi$

$f_{I_n} = \omega^2 m_n \phi_n = 100 \times 1 \times 1 = 100$

initially, $Q_n = f_I = 100$

$\Delta\phi_n = \frac{100}{600} = \frac{Q_n}{K_n} = 0.167$

Calculate. $\phi_{n-1} = \phi_n - \Delta\phi_n = 1 - 0.167 = 0.833$

$Q_{n-1} = f_{In} + f_{I(n-1)} = 100 + 100 \times 1.5 \times 0.833 = 224.95$

$\Delta\phi_{n-1} = \frac{Q_{n-1}}{K_{n-1}} = \frac{224.95}{1200} = 0.187$

$\phi_{n-2} = \phi_{n-1} - \Delta\phi_{n-1} = 0.833 - 0.187 = 0.646$

$f_{I(n-2)} = \omega^2 m_{(n-2)} v_{n-2} = 100 \times 2.0 \times 0.646 = 129.2$

$Q_{n-2} = P_{n-1} + f_{I(n-2)} = 224.95 + 129.2 = 354.15$

~~$v_0 = v_{n-2} - \Delta v_{n-2} = 0.646$~~

$\Delta v_{n-2} = \frac{Q_{n-2}}{k_{n-2}} = \frac{354.15}{1800} = 0.197$

$\therefore v_{n-3} = v_0 = v_{n-2} - \Delta v_{n-2} = 0.646 - 0.197 = 0.449$

| Trial – II $\omega^2 = 200$ | | | | Trial – III $\omega^2 = 209$ | | | |
|---|---|---|---|---|---|---|---|
| v | Δv | Q | $f_I$ | v | Δv | Q | $f_I$ |
| 1.0 | | | 200 | 1.0 | | | 209 |
| | 0.333 | 200 | | | 0.348 | 209 | |
| 0.667 | | | 200.1 | 0.652 | | | 204.402 |
| | 0.333 | 400.1 | | | 0.344 | 413.402 | |
| 0.334 | | | 133.6 | 0.308 | ~~0.38~~ | | 128.74 |
| | 0.296 | 533.7 | | | 0.301 | 542.14 | |
| 0.038 | | | | 0.007 | | | |





To have rough estimate we can interpolate linearly between $\omega_1$ & $\omega_2$.

$$\frac{200 - 100}{\omega - 100} = \frac{0.038 - 0.449}{0 - 0.449} \Rightarrow \omega = 209.24$$

Again interpolate.

$$\left(\frac{\Delta\omega^2}{\Delta v_B}\right)_{2-3} = \left(\frac{\Delta\omega^2}{\Delta v_B}\right)_{34}$$

$$\text{or} \quad \frac{9}{0.038} = \frac{\Delta\omega^2_{(34)}}{0.007} \Rightarrow \Delta\omega^2_{(34)} = 2.03$$

$\therefore \omega_1^2 = 211 \Rightarrow \omega_1 = 14.51$ rad/sec.

THANK YOU



3.10 Support excitation (Influence vector, synchronous support motion of a planar system and structure with multiple support motions)

THANK YOU